$PEACE

Tokenomics & TGE Master Report

Confidential internal & partner review

November, 2025

Final (Draft)

Executive Summary

Highlights

- **Design Objective:** Align token value with verifiable human impact and real commerce, eliminating discretionary emissions and hype-driven supply.

- **Fixed Supply & Mechanized Scarcity:** Max supply 10,000,000,000; claim-time burn of **0.75%** on verified rewards; surplus-only buybacks executed under strict market-hygiene guards (TWAP **7,200s**, quote freshness ≤ **120s**, deviation fences).

- **Use & Utility:** Payments and discounts within the Peace Store (policy-bounded), governance with a **0.1% per-identity vote cap**, and HII-verified claims that tie issuance to measurable outcomes.

- **Market Path:** Solana-native TGE via **Jupiter LFG**; primary liquidity on **Meteora DLMM** with mirrored CLMM pools; **LP locks ≥ 12 months** posted on-chain.

- **Distribution Shape:** Community & Ecosystem **45%**, Treasury **25%**, Team **15%** (12-month cliff; 24–36-month linear), Partners **10%** (milestone-gated), Liquidity **5%** (locked).

- **Governance & Controls:** Classed quorums (**10/30/60%**) with timelocks (**12/48/96h**), identity-bounded voting, program-enforced bounds, and universal receipts for auditability.

- **Transparency:** Monthly signed JSON manifests publish parameters, KPIs, and action receipts, enabling third parties to recompute runway, buybacks, and supply changes without trusted intermediaries.

Thesis. Most "impact" tokens degrade under wash activity and inflationary issuance. Peace Soldiers addresses this by issuing only against HII-verified actions and closing the loop through the Peace Store, where fees create surplus that can trigger buybacks subject to non-bypassable guards. Scarcity is a consequence of measurable activity, not narrative.

Mechanics That Withstand Adverse Conditions. Monetary policy is fail-closed. If freshness, deviation, or TWAP checks fail, buybacks do not execute. Coverage-Months floors prioritize solvency over optics; emission throttles engage automatically if runway falls below policy thresholds. Distribution and vesting are escrowed, milestone-gated, and paced by DEX absorption capacity to avoid unlock overhang.

Microstructure & Liquidity Quality. Concentrated, locked liquidity targets tight realized spreads and sufficient ±1% depth using a minimal token budget. LP locks are on-chain and immutable without constitutional governance, reducing “liquidity rug” risk and institutional risk premia.

Governance Legitimacy. Identity-bounded voting caps, community-major float, and classed quorums make capture economically and operationally infeasible. All privileged actions traverse a Governance Gateway that enforces bounds, timelocks, payload hashes, and receipt emission.

Risk Posture. Unified risk registers and guardrails address oracle failure, liquidity shocks, unlock pressure, governance apathy, fraud rings, and compliance shifts. Incident playbooks, external audits, reproducible builds, and MPC/HSM custody minimize operational and security risks.

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1.0 Quick Parameter Reference Annex A: Parameter Registry

Surface	Parameter	Value / Policy	Notes
Supply	Max Supply	**10,000,000,000 (fixed)**	Mint authority frozen at genesis.
Burn	Claim-Time Burn	**0.75%** of eligible reward at claim	Ties scarcity to verified impact.
Buybacks	Funding Source	Surplus only	No discretionary or deficit buybacks.
Buybacks	Execution Guards	**TWAP 7,200s; freshness ≤120s; deviation fences**	Fail-closed: non-compliant conditions abort.
Governanc e	Per-Identity Vote Cap	**0.1%**	Identity-aware voting; mitigates capture.
Governanc e	Quorums / Timelocks	**10/30/60%** with **12/48/96h**	Higher class for constitutional changes.
Liquidity	LP Locks	**≥12 months**	On-chain, immutable without governance.
Reporting	Manifests	Signed monthly JSON	Parameters, KPIs, receipts, changes.

1.1 Purpose and Value Thesis

The token functions as a coordination asset that prices access, participation, and policy within the Peace Soldiers economy. The design aligns scarcity with verifiable human impact and real commerce, replacing discretionary issuance with mechanized rules that fail closed under degraded market conditions.

1.1.1 Peer Benchmarks Contextual Comparison

Dimension	Peace Soldiers ($PEACE)	Uniswap (Fee Accrual)	Aave (Risk-Managed Incentives)	KlimaDAO (Rebasing/Impact)
Supply Model	Fixed supply; action-coupled claim-time burn; surplus-only buyback-and-burn	No native token burn; value via fee switch/treasury flows	Emissions oriented to risk/usage; staking-based security	Elastic supply via rebasing; inflation risk if demand lags
Value Accrual	Clean GMV → surplus → buybacks under hygiene guards	Trading fees; potential governance-gated captures	Interest spreads; safety module incentives	Carbon market exposure; protocol-owned liquidity
Governance	Identity-bounded voting caps; classed quorums/timelocks	Token-weighted; proposals/treasury control	Token-weighted; risk committees and parameters	Token-weighted; market-dependent policy
Risk Posture	Fail-closed buybacks; runway floors; receipts everywhere	Market-driven; treasury/fee decisions	Formal risk framework and caps	Macro/market cycles, carbon pricing volatility
Investor Takeaway	Scarcity tied to measurable activity; manipulation-resistant execution	Fee engine with governance optionality	Risk-priced incentives; security via staking	Mission-aligned but inflation-exposed in drawdowns

1.2 Utilities and Rights

- **Payments and Discounts:** Settlement and policy-bounded discounts across the Peace Store and integrated services.
- **Governance:** Proposal and voting rights subject to identity caps, classed quorums, and timelocks.

- **Impact Incentives:** Eligibility to claim rewards from HII-verified missions; all claims require verifiable receipts.

1.3 Supply Policy and Monetary Design

- **Fixed Ceiling:** 10B units; mint authority frozen at genesis.
- **Mechanized Scarcity:** Claim-time burn (0.75%) and surplus-only buyback-and-burn.
- **Fail-Closed Controller:** Buybacks execute only when TWAP, freshness, and deviation checks pass; otherwise, no action occurs.

Supply Identity. Let StS_tSt be circulating supply at epoch ttt:

St+1=St+Mt−Btclaim−BtbbS_{t+1} = S_t + M_t - B^{\text{claim}}_t - B^{\text{bb}}_tSt+1=St+Mt−Btclaim−Btbb

with Btclaim=0.0075×RtB^{\text{claim}}_t = 0.0075 \times R_tBtclaim=0.0075×Rt (eligible reward RtR_tRt) and BtbbB^{\text{bb}}_tBtbb sized from surplus subject to hygiene guards.

Scenario Table — Supply Trajectory (Visual).

Scenario	GMV Growth (y/y)	Claim Volume Index	Burn Yield (annual, %)	Buyback Cadence (per mo.)	Net Supply Change (annual, %)
Bear	10%	0.8×	0.6–1.0	0–1	~0% to −1%
Base	20%	1.0×	1.5–2.5	1–3	−2% to −4%
Hypergrowth	50%	1.8×	3.0–5.0	3–6	−4% to −7%

1.4 Sources and Sinks - Token Flow

- **Sources:** HII-verified mission rewards, milestone-gated partner grants, vesting streams.
- **Sinks:** Claim-time burn (0.75%), buyback-and-burn (surplus-only), policy/processing fees.

- **Closed-Loop Narrative:** Missions → Verified Receipts → Reward Claim (−0.75%) → Circulation → Store/Services → Surplus → Guarded Buybacks → Burn.

Components (visual Mix).

Component	Share of Surplus	Notes
Marketplace/Store Fees	45–60%	Policy-bounded discounts apply.
Premium Services	20–35%	Data/analytics/API tiers.
Misc. Program Revenues	10–20%	Grants, partnerships, yield on reserves.

1.5 Value Accrual

Accrual is tied to Clean GMV and verified outcomes. Claim-time burns reduce supply in proportion to realized impact, while surplus-only buybacks permanently remove supply when market hygiene permits. The mechanism avoids promises and relies on rules that convert real activity into durable scarcity.

1.6 Incentive Design-HII-Linked

- **Signal Ingestion:** Geotagged media, third-party attestations, and on-chain confirmations as per Receipt Schemas.
- **Scoring:** Submissions must exceed HII thresholds before eligibility unlocks; anomalous patterns are rejected or sampled for audit.
- **Anti-Fraud:** Rate limits, anomaly detection, and ring-fencing for collusion.

Evidence Weights and Quality (Visual).

Evidence Class	Weight	False-Positive Risk	Handling
On-Chain Attestation	High	Low	Direct eligibility trigger
Geotagged Media	Medium-High	Medium	Cross-check with metadata
Third-Party NGO Receipt	High	Low-Medium	Sampled audit, revocation path
Manual Report	Low	High	Requires corroboration

1.7 Treasury Policy and Runway

- **Coverage-Months (CM) Floors:** Treasury maintains CM above policy floors; emission throttles engage below thresholds.
- **Surplus Allocation:** Surplus above CM floors may fund buybacks; otherwise retained as reserves.
- **Execution Discipline:** Hygiene guards and pacing prevent adverse feedback loops.

Runway States (Visual).

CM State	CM Range	Policy Response
Stable	≥ 18m	Normal operations; surplus eligible for buybacks under guards
Caution	12–18m	Reduce buyback pacing; prioritize reserves
Defense	< 12m	Suspend buybacks; throttle emissions; publish remediation plan

1.8 KPIs and Operating Targets-Dashboard

KPI	Definition	Current Trend	Target
Clean GMV	GMV net of wash/manipulation	▲ improving	Upward trajectory
Burn Yield	(Bclaim+Bbb)/S(B^{\text{claim}} + B^{\text{bb}})/S(Bclaim+Bbb)/S	● stable	Rising with adoption
UPI(30)	Unique paying identities (30d)	▲ improving	Consistent growth
DEX Depth ±1%	Aggregated depth at ±1%	● stable	Sufficient for routine flow
FFR	Freshness-fail rate	● low	Near-zero
CM	Coverage-Months runway	● at floor	≥ floor across cycles

Legend: ▲ improving · ● stable

1.9 Modeling and Quantitative Scenarios

Assumptions (Visual):

GMV growth 10/20/50% (bear/base/hypergrowth); surplus margin 8–15%; controller passes hygiene in 40–70% of epochs depending on volatility.

Outputs.

Scenario	Avg. Burn Yield (annual)	Buyback Frequency (per mo.)	CM Stability	Notes
Bear	0.6–1.0%	0–1	At/above floor	Controller mostly inactive; reserves preserved
Base	1.5–2.5%	1–3	Above floor	Periodic buybacks; steady claim-time burns
Hypergrowth	3.0–5.0%	3–6	Above floor	Frequent buybacks; guards cap cadence

Sensitivity analyses in Annex B parameterize outcomes by GMV growth, mission mix, λ-decay in scoring, and market-hygiene thresholds.

1.10 Risks and Mitigations

Token-Design Specific, Scored

Risk	Likelihood	Impact	Mitigation	Residual
Oracle / Market Hygiene Failure	Low-Med	High	TWAP/freshness/deviation guards; fail-closed; external monitoring	Low-Med
Wash / Collusion	Medium	Medium	HII thresholds; anomaly detection; rate limits; audit sampling	Low-Med
Governance Capture	Low-Med	High	Identity caps; classed quorums/timelocks; constitutional gating	Low
Liquidity Shock	Medium	Medium-High	Locked primary LP; mirrored pools; band engineering; pacing	Low-Med
Unlock Overhang	Low-Med	Medium	Cliffs/linears; milestone tranches; absorption tests	Low
Compliance Shift	Medium	High	Mode switches; geofencing; elevated governance thresholds	Med

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1.11 Long-Term Equilibrium and Sustainability

The design targets a steady-state where Clean GMV growth sustains a modest, persistent deflationary pressure (illustratively **2–5% annual** net reduction during mature phases). Claim-time burns scale with realized impact; surplus-only buybacks convert operational value into permanent supply removal under guardrails. Equilibrium is maintained by CM floors, execution pacing, and identity-bounded governance that reduces incentives for extractive policy changes.

1.12 Auditability and Receipts Machine-Verifiable

All privileged and monetary actions emit signed receipts. Example minimal receipt (illustrative schema; see Annex D):


```
{
  "action": "buyback_execute",
  "epoch": 12345,
  "twap_window_s": 7200,
  "freshness_s": 48,
  "deviation_ok": true,
  "notional_usd": 125000.00,
  "base_token": "PEACE",
  "counter_token": "USDC",
  "tx_hash": "5y...Zk",
  "controller_version": "bb-1.3.2",
  "policy_ref": "AnnexA.v1.4",
  "signer": "MPC:PS-TREASURY-01"
}
```


Section 2

Distribution and Vesting

2.0 Allocation Summary and Quick Parameter Reference

Bucket	Allocation	Vesting / Cliff	Release Mode	Primary Rationale
Community and Ecosystem	**45%**	Programmatic; claim-gated	Missions, grants, curated programs	Broad float, usage growth, governance legitimacy
Treasury (Strategic)	**25%**	N/A (reserve)	Discretionary under policy bounds	Runway resilience; partner co-funding; buyback eligibility under guards

Team	**15%**	12-month cliff; 24–36-month linear	Linear post-cliff	Long-term alignment; anti-churn retention
Partners	**10%**	Milestone-gated	Tranche unlocks upon verification	Outcome-linked integrations; value-for-value delivery
Liquidity	**5%**	LP locked ≥12 months	Seed + mirrored pools	Depth, tight spreads, durable price discovery
Total	**100%**	—	—	—

2.1 Allocation Buckets and Rationale

Allocation balances decentralization, market quality, and solvency. Community-major float supports governance breadth and usage incentives; treasury reserves maintain coverage-months across cycles; team and partner tranches vest on timelines consistent with durable delivery.

2.1.1 Market Benchmarks

Dimension	**Indicative Market Range (2024–2025)**	**PS Positioning**	**Commentary**
Community / Ecosystem	~40–45%	**45%**	Tilted to breadth and usage incentives
Treasury / Reserves	~20–25%	**25%**	Within norm; supports runway and co-funding

Core Team	~12–20%	**15%**	Mid-range; reduces concentration risk
Investors / Partners	~10–20%	**10%**	Lower bound; milestone-linked to value creation
Liquidity	~3–7%	**5%**	Sufficient for early depth with locks

Note: Ranges reflect aggregated public tokenomics surveys and launch reports; exact sources enumerated in the bibliography/Annex references.

2.1.2 Vesting Schedule Types

Type	**Pros**	**Cons**	**Usage in PS**
Linear	Smooth supply profile; predictable	Less performance-sensitive	Team post-cliff
Cliff + Linear	Strong retention gate; reduces early sell-pressure	Hard unlock at cliff	Team 12-month cliff, then linear
Graded (stepped)	Operational simplicity	Step-wise supply shocks	Select grants only
Milestone-Based	Outcome-aligned; mitigates underperformance	Verification overhead	Partners, integrations, BD

Hybrid (Milestone + Linear)	Balance of predictability and performance	Complexity	Large strategic partners

Milestone formula (partners). For tranche iii in total NNN milestones, with verified completion set MMM at time ttt:

releasedt = V · ∣M∣N · min (1,t−tfirst_milestoneP)\text{released}_t \;=\; V \cdot \frac{|M|}{N}\cdot \min\!\left(1,\frac{t-t_{\text{first_milestone}}}{P}\right)releasedt=V · N∣M∣ · min(1,Pt−tfirst_milestone)

Unmet milestones defer release; unverified claims do not accrue.

2.2 Vesting Mechanics and Escrows

- **Team:** 12-month cliff; 24–36-month linear vest. No TGE issuance. No acceleration absent constitutional change.
- **Partners:** Milestone-gated tranches; independent verification; clawback/expiry clauses for non-delivery.
- **Community Programs:** Emissions route through claim receipts; claim-time burn (0.75%) applies at reward redemption.
- **Liquidity:** Seeded at TGE; LP tokens locked ≥12 months in program-controlled custody.

Smart-contract implementation notes.
Audited escrow contracts with immutable parameters (`start_ts`, `cliff_ts`, `period_s`, `rate`, `beneficiary`) reduce manipulation risk. All unlocks emit machine-verifiable receipts (see 2.6).

2.3 Unlock Calendar, Absorption, and Sensitivities

Unlocks are published at genesis and updated via signed monthly manifests. Each epoch applies release caps to respect market capacity.

Absorption caps (per epoch ttt).

At ≤ min(κ · ADV30,t, ρ · D±1%,t, Utschedule)A_t \;\le\; \min\Big(\kappa \cdot \text{ADV}_{30,t},\;\rho \cdot D_{\pm 1\%,t},\;U^{\text{schedule}}_t\Big)At≤min(κ · ADV30,t,ρ · D±1%,t,Utschedule)

- **ADV cap κ\kappaκ** (illustrative): 0.05 (≤5% of 30-day ADV).
- **Depth cap ρ\rhoρ** (illustrative): 0.20 (≤20% of aggregated ±1% depth).
- **Volatility gate:** If realized volatility exceeds policy threshold, At=0A_t = 0At=0 (defer and carry-forward).

Sensitivity table

Market State	**Volatility**	**κ\kappa κ**	**ρ\rho ρ**	**Released**	**Carry-Forward**	**UPI Impact**
Bear	High	0.03	0.10	Low	High	Neutral–Positive (stability preserved)
Base	Medium	0.05	0.20	Medium	Low	Positive (predictable flow)
Hypergrowth	Low–Med	0.07	0.25	High	Low	Positive (demand absorbs supply)

Visuals will be added to the final report

- **Fig. 2-A:** Multi-year Gantt of unlocks by bucket with overlay of ADV/depth caps and volatility gates.

- **Fig. 2-B:** Cumulative unlocks vs. capacity; shaded deferral bands.

2.4 Addresses, Timelocks, and LP Locks

Program-controlled vaults isolate buckets; timelocked governance enforces payload hashes and bound checks; primary LP locks ≥12 months; modifications require constitutional action.

2.5 Fairness, Anti-Concentration, and Sybil Resistance

- **Identity caps:** Per-identity participation limits in community programs.
- **Tiered emissions:** If concentration metrics breach thresholds, emissions tilt toward breadth-expanding routes.
- **Sybil controls:** Wallet clustering heuristics, device/behavioral signals, optional KYC modes; anomalous clusters ring-fenced.
- **Targets:** Gini coefficient < 0.60; Top-10 share ≤ γ=25%\gamma=25\%γ=25% sustained.

Example. If Top-10 share >γ>\gamma>γ for KKK consecutive epochs, redirect **10%** of forthcoming community emissions to breadth programs until metrics normalize.

Visuals will be added to the final report

- **Fig. 2-C:** Holder deciles bar chart and Gini trend lines across scenarios.

2.6 Operational Controls and Receipts

Every release action emits a signed JSON receipt with schedule identifiers, caps applied, carry-forward, and transaction hashes. Receipts enable third parties to reconcile circulating supply against the unlock calendar.

Example

```json
{
  "action": "unlock_release",
  "epoch": 12402,
  "bucket": "Partners",
  "schedule_id": "PART-2025-05-A",
  "calendar_due": 5_000_000,
  "released": 3_500_000,
  "caps": {"adv": true, "depth": true, "vol_gate": false},
  "carry_forward": 1_500_000,
  "tx_hash": "9K...3s",
  "policy_ref": "AnnexA.v1.4",
  "signer": "MPC:PS-VEST-01"
}
```

2.7 Monitoring and Risk Framework KPIs with Linked Mitigations

Metric (KPI)	Definition	Guard / Target	Linked Risk	Triggered Mitigation
Float Breadth	Addresses ≥ threshold	+15% QoQ	Concentration	Redirect 10% emissions to breadth programs
Top-10 Share	% of supply held by top 10	≤ γ=25%\gamma =25\%γ=25%	Capture / Dump Risk	Tiered emission reductions; partner deferrals
Gini Coefficient	Inequality measure	< 0.60	Concentration	Broaden community airdrop eligibility

Unlock Drift		Deviation from calendar	≤ 5%	Execution / Liquidity
Depth ±1%	Aggregated band depth	≥ policy floor	Liquidity Shock	Reinforce LP band; pause releases
ADV Ratio	30d ADV / float	Stable↑	Absorption Risk	Lower κ\kappaκ; reschedule
UPI(30)	Unique paying identities	Upward trend	Adoption	Retarget community grants
Incident Count	Release-related incidents	0	Operational Risk	Freeze; audit; corrective vote

Visuals will be added to the final report

- **Fig. 2-D:** Risk matrix heatmap (likelihood × impact) with residual after mitigations.

2.8 Peer Practices and Case References

Project	Model Element	Practice	Relevance to PS
Uniswap	Community incentives / fee engine	Broad community allocation and governance control of fees	Supports community-major thesis

Aave	Risk-managed incentives	Staking and safety-module risk pricing	Informs guardrails and treasury posture
Yearn	Fair-launch dynamics	No pre-allocations; high volatility	Highlights need for vesting and absorption tests
EigenLayer	Milestone-linked rewards	Outcome-gated releases	Mirrors partner tranche design
KlimaDAO	Impact-linked design	Market-cycle sensitivity	Reinforces fixed supply and mechanized scarcity choice

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2.9 Equilibrium Modeling

Assumptions: baseline 20% annual Clean GMV growth; treasury surplus margin 8–15%; identity measures maintain Gini < 0.60; controller passes hygiene in 50–65% of epochs.

Outcomes: vesting completes without sustained concentration drift; stock-to-flow supports modest deflationary pressure; unlock deferrals in high-volatility windows keep realized spreads within target bands; community breadth and UPI growth remain positively correlated with distribution cadence.

2.10 Visual Index (*will be added to the final report*)

- **Fig. 2-A:** Multi-year unlock Gantt with caps and gates
- **Fig. 2-B:** Cumulative unlocks vs. capacity (deferral shading)
- **Fig. 2-C:** Ownership concentration dashboard (deciles, Gini, Top-10)
- **Fig. 2-D:** Risk matrix heatmap (likelihood × impact, residual)

Section 3

TGE Mechanics and Liquidity

3.0 Design Principles

- **Fair Discovery:** Batch-based price discovery reduces gas/MEV advantages and front-running.
- **Market Quality First:** Liquidity seeded to target tight realized spreads and sufficient ±1% depth using minimal token budget.
- **Guarded Execution:** All sale and market-making actions pass machine-checkable hygiene tests; fail-closed by default.
- **Transparency:** Pre-announced parameters, on-chain receipts, and a publish-once unlock calendar.

3.1 Objectives and Constraints

Objectives

1. Discover a credible clearing price with broad participation and minimal slippage at launch.
2. Establish durable primary liquidity (locked) and mirrored depth on major venues.
3. Minimize volatility and unlock overhang during the first 90 days.

Constraints

- No discretionary buybacks pre-TGE; no team unlocks at TGE.
- Regional compliance modes enforced at the participation gateway.
- Liquidity operations must respect TWAP, freshness, and deviation fences.

3.2 Launch Architecture and Venues

- **Discovery Venue:** Batch auction / uniform-price style sale (e.g., launch aggregator or auction portal) with a fixed participation window and per-identity caps.

- **Primary Liquidity Venue:** Programmable concentrated AMM with dynamic bins/bands (e.g., DLMM/CLMM).

- **Mirrors:** Secondary pools on major DEXs to reduce fragmentation risk; routing compatibility ensured via aggregators.

Operational Flow

1. **Preparation:** Publish sale doc, addresses, auction parameters, and K-factors (caps, identity mode, anti-bot list).

2. **Discovery:** Accept commitments during the auction window; compute **uniform clearing price** p*p^*p* post-window.

3. **Settlement:** Allocate tokens pro-rata at p*p^*p*; refund surplus capital.

4. **Liquidity Seed:** Mint liquidity seed to custody vault; deploy bands around p*p^*p* and lock LP.

5. **Mirrors:** Seed mirrored pools with proportionally thinner bands; register with routers.

3.3 Price Discovery Mechanism

Uniform-Price Auction (UPA).

Let bids be {(qi,bi)}\{(q_i, b_i)\}{(qi,bi)} where qiq_iqi is quantity requested and bib_ibi the bid price. Aggregate demand Q(p)Q(p)Q(p) declines with ppp. Clearing price p*p^*p* satisfies:

Q(p*) = SsaleQ(p^*) \;=\; S_{\text{sale}}Q(p*)=Ssale

All filled participants pay p*p^*p*. Pro-rata allocation applies when the last filled tranche is over-subscribed.

Anti-Manipulation Guards

- Minimum unique participant count UminU_{\min}Umin.
- Per-identity cap qmaxq_{\max}qmax.
- Bid freshness and funding validity checks.
- Disqualification for known Sybil clusters (pre-computed allowlist denial set).

Outputs Published

- p*p^*p*, total participants, oversubscription ratio, effective float at TGE, and receipt hashes.

3.4 Participation Rules and Compliance Modes

- **Identity Modes:**
  - *Open Mode* (default): Light identity and Sybil heuristics; per-identity caps.
  - *KYC Mode* (jurisdictional): Full KYC for restricted regions; geographic gating.
- **Per-Identity Caps:** Enforced on commitments and allocations.
- **Funding Rules:** On-chain stablecoins or native SOL, escrowed during the window.
- **Sanctions/Denied Lists:** Enforced at gateway; appeals via governance queue (non-blocking to TGE).

3.5 Anti-Bot and Fairness Controls

- **Time-Sliced Buckets:** The auction window is discretized into sub-windows; identical pricing; prevents last-second sniping optics.
- **Commit-Reveal Option:** Optional hashed bids; reveal window after commit phase.
- **Velocity Limits:** Caps on per-address order change frequency; throttle transactions from shared infrastructure fingerprints.
- **Router Neutrality:** No privileged routers; aggregator parity guaranteed.

3.6 Circulating Set at TGE

Initial circulating supply C0C_0C0 is the sum of sale allocations settled, liquidity seed, and any programmatic incentives that start at TGE:

C0 = Ssale + Lseed + Eprog,0C_0 \;=\; S_{\text{sale}} \;+\; L_{\text{seed}} \;+\; E_{\text{prog},0}C0=Ssale+Lseed+Eprog,0

with SsaleS_{\text{sale}}Ssale the distributed sale tokens, LseedL_{\text{seed}}Lseed the LP seed minted to the custody vault, and Eprog,0E_{\text{prog},0}Eprog,0 restricted to small, claim-gated incentives if enabled. Team and partner allocations remain escrowed (Section 2).

Reporting at TGE

- C0C_0C0, p*p^*p*, total raise, participant count, LP seed notional and band map, mirror pool IDs, and LP lock receipts.

3.7 Liquidity Provisioning and Band Engineering

Goals

- Realized spread within target bps.
- Sufficient depth at ±1% around p*p^*p*.
- Controlled inventory risk under volatility shocks.

Band Map (Visual)

Band	Price Range (relative to p*p^*p*)	Target Depth Share	Notes
Core	[−0.5%,+0.5%][-0.5\%, +0.5\%][−0.5%,+0.5%]	55–65%	Primary trading band
Near	[−1.0%,−0.5%]∪[0.5%,1.0%][-1.0\%, -0.5\%] \cup [0.5\%, 1.0\%][−1.0%,−0.5%]∪[0.5%,1.0%]	25–35%	Shock absorber
Guard	Outside ±1.0%	5–15%	Tail defense; thinner density

Deployment Rules

- **TWAP Anchor:** Price oracle uses rolling TWAP; band centers follow TWAP with dampening.
- **Freshness/Deviation Fences:** If freshness >>> threshold or deviation >>> fence, rebalance is deferred.
- **Locking:** Primary LP tokens locked ≥12 months; mirrors use shorter locks with policy-bounded rebalances.

Depth / Slippage Targets

For trade notional xxx and mid-price mmm, target slippage σ(x)\sigma(x)σ(x) within band width www:

σ(x) ≈ xD±wwithD±w=depth at ±w\sigma(x) \;\approx\; \frac{x}{D_{\pm w}} \quad\text{with}\quad D_{\pm w} = \text{depth at } \pm wσ(x)≈D±wxwithD±w=depth at ±w

Design sets D±1%D_{\pm 1\%}D±1% to satisfy routine order flow with σ(x)\sigma(x)σ(x) below policy bps.

3.8 Use of Proceeds and Treasury Posture

- **Priority:** Coverage-Months (CM) floors before any optional buyback eligibility in later phases.
- **Allocation Buckets:** Engineering, security and audits, ecosystem integrations, liquidity operations, and operations runway.
- **Controls:** Disbursements require governance-class authorization; manifests publish notional changes and remaining runway.

3.9 Post-TGE Stabilization Runbook

Phase A — Discovery to Listing (D+0 to D+3)

- Publish receipts, addresses, and LP lock confirmations.

- Activate mirrors; verify router paths; monitor spreads and depth.

Phase B — Early Stabilization (D+4 to D+21)

- Rebalance bands if TWAP drift within policy; otherwise defer.
- Enforce volatility gates on any scheduled emissions (if any).
- Begin monthly KPI reporting cadence.

Phase C — Consolidation (D+22 to D+90)

- Gradual normalization of band density as volume stabilizes.
- No discretionary buybacks; controller remains inactive unless policy permits in later phases.
- Incident-response readiness and external monitoring on oracles, depth, and volatility.

3.10 Risks and Contingencies TGE/Liquidity

Risk	**Description**	**Likelihood**	**Impact**	**Mitigation**	**Residual**
Auction Manipulation	Collusion or bid spoofing	Low–Med	Medium	UPA with commit-reveal; per-identity caps; minimum UminU_{\min}Umin	Low
Volatility Shock	Large price swings at listing	Medium	High	Wide-to-tight band strategy; mirrors; volatility gates	Low–Med
Oracle Drift / Stale Quotes	Mis-anchored rebalances	Low–Med	High	Freshness and deviation fences; external monitors	Low
Liquidity Fragmentation	Depth split across venues	Medium	Medium	Mirrored pools and router coordination	Low–Med
MEV / Sniping	Edge from latency and routing	Medium	Medium	Batch settlement; time-sliced windows; router neutrality	Low–Med

Compliance Event	Sudden regional restriction	Low–Med	High	Mode switch to KYC; geo-gating; governance notification	Med

3.11 Metrics and Alerts First 90 Days

Metric	**Definition**	**Target / Guard**	**Alert**
Realized Spread	VWAP-mid spread over intervals	Within policy bps	Breach → widen core band density
Depth ±1%	Aggregated depth at ±1%	≥ floor	Breach → reinforce near bands
Volatility	Realized σ\sigmaσ over rolling window	≤ threshold	Breach → pause rebalances/emissions
Price Freshness	Oracle quote age	≤ freshness cap	Breach → fail-closed controller
ADV Ratio	30d ADV / float	Stable↑	Drop → defer unlocks
Incident Count	Liquidity/oracle incidents	0	>0 → freeze and publish RCAs

3.12 Visual Index (will be injected to the final report)

- **Fig. 3-A:** Auction demand curve and clearing price p*p^*p*.
- **Fig. 3-B:** LP band map (core/near/guard) around p*p^*p*.
- **Fig. 3-C:** Cumulative volume vs. realized spread (D+0…D+21).
- **Fig. 3-D:** Depth at ±1% across venues and time.
- **Fig. 3-E:** Post-TGE stabilization timeline (Gantt).

Section 4

Governance and Controls

4.0 Design Principles

- **Legitimacy:** Identity-bounded participation prevents plutocratic capture while preserving economic stake signals.
- **Predictability:** Classed decisions, fixed quorums, and deterministic timelocks reduce discretionary risk.
- **Safety by Construction:** Parameter bounds and automated checks enforce fail-closed execution.
- **Auditability:** Every governance transition emits machine-verifiable receipts; proposal payloads are hash-committed.

4.1 Governance Classes and Scope

Class	Scope	Examples	Quorum	Timelock
C1 — Routine	Operational and informational actions within pre-set bounds	Publishing manifests; rotating non-privileged keys; enabling a pre-audited view	**10%**	**12h**
C2 — Parameterized	Changes to bounded parameters and policy knobs	Adjusting TWAP window, freshness cap, liquidity band density, unlock caps	**30%**	**48h**

C3 — Constitutional	Foundational changes	Supply rules, governance cap, contract upgrades, LP lock policy, compliance modes	**60%**	**96h**

Notes:

- Boundaries for C2 are codified in Annex A (Parameter Registry).
- Any proposal that would breach a bound is rejected at the gateway pre-vote.

4.2 Voting Mechanics and Identity Cap

Eligibility and Identity.

Voters are unique identities (Passport-verified). Token holdings signal stake but are capped per identity to curb concentration.

Per-Identity Cap.

Let SSS be total supply. The maximum counted voting weight per identity is:

w_{\max} \;=\; 0.001 \times S \quad (\text{0.1% cap})

Actual voting weight for identity jjj with stake sjs_jsj is:

wj = min(sj, wmax)w_j \;=\; \min(s_j,\; w_{\max})wj=min(sj,wmax)

Quorum and Outcome.

For a proposal of class ccc with quorum QcQ_cQc, total counted weight WWW, and “For” weight FFF:

WS≥QcandFW>12\frac{W}{S} \ge Q_c \quad\text{and}\quad \frac{F}{W} > \tfrac{1}{2}SW≥QcandWF>21

C3 proposals may require supermajority thresholds on specific actions (e.g., 2/3).

Delegation.

Optional delegation is permitted with (i) the per-identity cap applied to the delegate after aggregation, and (ii) **auto-expiry** each quarter unless renewed. Delegation graphs are acyclic; maximum inbound delegates per identity may be bounded (policy surface).

Proposal Rights.

Proposal creation requires a minimum support signal (e.g., endorsements from nnn distinct identities or a fee-backed deposit) to discourage spam.

4.3 Quorums, Timelocks, and Emergency Handling

- **Fixed Timelocks:**

  Class-specific delays (12/48/96h) ensure transparent review windows.

- **Staging:**

  Multi-step execution for complex payloads (approve → stage → execute) with hash-locked payloads.

- **Emergency Pause:**

  A narrowly scoped, time-boxed pause contract can halt **new** privileged actions for up to TmaxT_{\max}Tmax (e.g., 24h) upon objective triggers (e.g., oracle failure). Activation itself is a C2 action bound by receipts and auto-expiry.

4.4 Proposal Lifecycle and Receipts

States:

Draft → Gate Check → Vote Open → Vote Close → Timelock → Execute → Receipt → Post-Execution Verification.

Gateway (pre-vote) checks:

- Parameter bounds (Annex A)
- Payload hash and schema validity
- Identity counts and duplicate suppression
- Compliance mode compatibility (if applicable)

Execution receipts:


```
{
  "action": "gov_execute",
  "proposal_id": "C2-2025-045",
  "class": "C2",
  "payload_hash": "0x…",
  "quorum": 0.30,
  "for_weight": 98500000,
  "against_weight": 14500000,
  "abstain_weight": 2500000,
  "exec_block": 21945012,
  "signer": "MPC:PS-GOV-01",
  "policy_ref": "AnnexA.v1.4"
}
```


4.5 Parameter Surfaces Bounded Knobs

Surface	Parameter	Class	Bound / Range	Default
Market Hygiene	TWAP window (s)	C2	3,600–14,400	**7,200**
Market Hygiene	Freshness cap (s)	C2	10–180	**120**
Market Hygiene	Deviation fence (%)	C2	0.25–2.00	**0.75**
Liquidity	Core band share (%)	C2	40–75	**60**

Liquidity	Near band share (%)	C2	15–45	**30**
Distribution	ADV cap κ\kappaκ	C2	0.02–0.10	**0.05**
Distribution	Depth cap ρ\rhoρ	C2	0.10–0.30	**0.20**
Treasury	Coverage-Months floor	C2	9–24	**18**
Governance	Per-identity cap (% of S)	C3	0.05–0.20	**0.10**
Governance	Delegation expiry (days)	C2	30–180	**90**
Compliance	Mode flag (Open/KYC)	C3	Enum	**Open**

Any parameter outside ranges requires C3 to modify bounds first.

4.6 Constitutional Changes

Constitutional scope includes: supply policy, identity cap, governance class definitions, contract upgrade routes, LP lock policy, and compliance modes. C3 proposals entail:

- **Supermajority and Extended Timelock:**

  ≥60% quorum with ≥2/3 approval where specified; 96h timelock.

- **Two-Epoch Confirmation:**

  Optional second-epoch confirmation for irreversible actions.

- **Rollback Route:**

If post-execution checks fail, a predefined rollback payload can be executed within a limited window.

4.7 Delegation, Turnout, and Apathy Controls

- **Auto-Expiry Delegation:**

  Delegations lapse every quarter unless renewed.

- **Turnout Incentives (non-emissive):**

  Reputation points attached to governance identities (off-token) for participation; no monetary yield.

- **Quorum Floors:**

  Class-specific floors prevent low-participation approvals.

- **Agenda Hygiene:**

  Proposal deposit (refundable if the proposal meets a minimum participation threshold).

4.8 Compliance Modes and Regional Constraints

- **Modes:**

  Open (default) and *KYC* (jurisdictional). Mode changes are constitutional.

- **Geo-Gating:**

  Jurisdiction logic executed at the gateway layer; proposals cannot bypass regional restrictions.

- **Records:**

  Compliance actions emit receipts for audit trails; access logs are hashed and time-stamped.

4.9 Monitoring and Governance KPIs

KPI	Definition	Target / Guard	Action on Breach
Participation Rate	W/SW/SW/S during voting windows	≥ quorum per class	Reduce proposal batching; extend comms window
Unique Voter Count	Distinct identities per epoch	Upward trend	Community outreach; adjust delegation expiry
Cap Breach Attempts	Proposals failing gateway cap checks	Near-zero	Educate proposers; enforce deposits
Execution Latency	Vote close → execution (h)	Within class timelock	Investigate bottlenecks; scale infrastructure
Revert/Failure Rate	Post-execution reverts (%)	Near-zero	Hardening reviews; raise class requirement
Compliance Exceptions	Mode-related denials	Near-zero	Clarify policy; governance notice

4.10 Risks and Mitigations Governance-Specific

Risk	Description	Likelihood	Impact	Mitigation	Residual

Vote Buying / Bribery	Off-chain incentives distort outcomes	Medium	Medium	Identity caps; delegation expiry; transparency dashboards	Low–Med
Governance Capture	Large holders coordinate within cap	Low–Med	High	0.1% cap; classed quorums/timelocks; supermajority for C3	Low
Voter Apathy	Low turnout leads to thin quorums	Medium	Medium	Deposits, auto-expiry delegation, comms cadence	Low–Med
Implementation Error	Faulty payloads or upgrades	Low	High	Hash-locked payloads; staged execution; rollback route	Low
Compliance Drift	Region-specific rules change	Medium	High	Mode switches; geo-gating; constitutional guard	Med
Identity Evasion	Sybil attempts to bypass cap	Medium	Medium	Multi-signal identity, clustering heuristics, periodic re-validation	Low–Med

4.11 Visual Index (*Visuals will be added to the final report*)

- **Fig. 4-A:** Governance class stack (C1/C2/C3) with scope, quorum, and timelock.
- **Fig. 4-B:** Voting pipeline: identity → cap → delegation → tally → threshold checks.
- **Fig. 4-C:** Parameter surfaces map with allowed ranges (Annex A overlays).

- **Fig. 4-D:** Proposal lifecycle sequence diagram with gateway checks and receipts.
- **Fig. 4-E:** Governance risk matrix (likelihood × impact, residual).

Section 5

HII and Mission Economics

5.0 Design Principles

- **Impact First:**

  Rewards are unlocked only for verifiable, high-confidence missions; issuance rejects unverifiable or low-signal submissions.

- **Defense in Depth:**

  Multi-signal evidence, anomaly scoring, rate-limits, and sampling audits minimize fraud and wash behavior.

- **Receipts Everywhere:**

  Each decision (accept/reject/partial) emits machine-readable receipts for third-party recomputation.

- **Privacy by Construction:**

  Signals favor privacy-preserving proofs and minimal disclosure consistent with integrity.

5.1 Quick Parameter Reference (Annex A)

Surface	Parameter	Policy / Range	Default	Notes
Eligibility Threshold	SminS_{\min}Smin (score)	Calibrated per template	**0.62**	Per-mission template ROC-tuned
Anomaly Gate	AmaxA_{\max}Amax	0.0–1.0	**0.35**	Above → auto-reject or escalate
Rate Limit	Claims per identity per window	1–5 per 24–168h	**2 / 72h**	Rolling credits model
Geo Consistency	Max drift vs. mission AOI	25–300 m	**75 m**	Template-specific
Device Uniqueness	Device re-use cooling	1–30 days	**7 days**	Per evidence class
Audit Sampling	Tier-1 sample rate	1–10%	**3%**	Adaptive: ↑ with drift
Dispute Window	Time to dispute decision	24–168h	**72h**	Tiered settlement
Burn on Claim	Claim-time burn	Fixed	**0.75%**	Applies on eligible reward

5.2 Signal Taxonomy and Evidence Ingestion

Evidence classes

are ingested with normalized quality scores $q_k \in [0,1]$ and weights w_k set per template:

- **On-chain attestations:** Programmatic confirmations, oracle proofs.
- **Geotagged media:** Timestamped images/video with sensor metadata.
- **Third-party NGO receipts:** Signed confirmations of delivery or completion.
- **Structured forms and IoT telemetry:** Water flow meters, pump uptime, classroom attendance sensors.
- **Manual narratives:** Low-weight; require corroboration.

Normalization pipeline

1. Schema validation → 2. Signature and metadata checks → 3. Cross-signal correlation (time/space/device) → 4. Privacy filters → 5. Scoring.

5.3 Scoring Model and Thresholds

Let $E = \{e_k\}$ denote accepted evidence items, q_k their quality, and w_k the template weights. Auxiliary consistency terms include geospatial consistency g, temporal consistency t, and device uniqueness u (each in $[0,1]$). A fraud-risk penalty $\phi \in [0,1]$ is learned from anomaly features.

$$\begin{aligned} S &= \Big(\sum_{k} w_k \, q_k \, \mathbb{1}[e_k \text{ passes validation}]\Big)\cdot G - \phi, \\ G &= \alpha_g g + \alpha_t t + \alpha_u u,\quad \alpha_g+\alpha_t+\alpha_u = 1. \end{aligned}$$

Decision rule

Eligible ⇔ S≥Smin ∧ A≤Amax\text{Eligible} \;\Leftrightarrow\; S \ge S_{\min} \;\wedge\; A \le A_{\max}Eligible⇔S≥Smin∧A≤Amax

where AAA is the anomaly score derived from clustering, duplicates, and out-of-distribution sensors.

Calibration

- Template-level ROC/PR curves generated on labeled samples.
- SminS_{\min}Smin chosen to target **precision ≥ 0.95** with recall optimized subject to budget.
- Thresholds versioned and referenced in receipts.

5.4 Reward Sizing and Monetary Hook

For a mission template jjj with base payout PjP_jPj and scoring multiplier m(S)m(S)m(S) (capped), the pre-burn reward is:

Rj = Pj · m(S),m(S)=min (1, β0+β1S).R_j \;=\; P_j \cdot m(S), \quad m(S) = \min\!\left(1,\;\beta_0 + \beta_1 S\right).Rj=Pj · m(S),m(S)=min(1,β0+β1S).

Claim-time burn applies:

Rjnet = Rj · (1−0.0075).R^{\text{net}}_j \;=\; R_j \cdot (1 - 0.0075).Rjnet=Rj · (1−0.0075).

Rewards never exceed budget ceilings per epoch; unfilled budget rolls forward.

5.5 Audit Framework and Sampling

- **Tier-0 (Automated):**

  Deterministic checks and model inference.

- **Tier-1 (Sampled Audit):**

Stratified 3% baseline; rate increases when drift occurs (e.g., area shifts, device reuse spikes).

- **Tier-2 (External Verification):**

  Third-party rechecks for high-value or high-risk clusters.

Metrics

- Precision/Recall on audited sets, False-Positive Rate (FPR), and **Audit-Fail Rate (AFR)** per template.
- Weekly calibration reports; thresholds adjusted via bounded governance.

5.6 Anti-Fraud Controls and De-Wash

- **Graph analytics:** Wallet/device clustering, shared infrastructure fingerprints, temporal stack analysis.
- **Geo-temporal plausibility:** Hard bounds on travel time and AOI radius.
- **Reputation decay:** Off-token reputation scores with decay; high fail rates lower future weights.
- **Ring-fencing:** Suspect clusters quarantined; claims moved to Tier-2 or denied.
- **Back-pressure:** Dynamic reduction of qkq_kqk weights for over-represented signals (e.g., identical EXIF patterns).

5.7 Eligibility, Rate Limits, and Throttles

Per-identity credits replenish over a rolling window; missions exceeding credits are queued. Let identity iii have credits ci(t)c_i(t)ci(t) and cap CCC:

ci(t+Δ) = min (C, ci(t)+λΔ),consume 1 per claim.c_i(t+\Delta) \;=\; \min\!\big(C,\; c_i(t) + \lambda \Delta\big),\quad \text{consume }1\text{ per claim}.ci(t+Δ)=min(C,ci(t)+λΔ),consume 1 per claim.

- **Defaults:** C=2C=2C=2 over 72h; λ\lambdaλ tuned to regain 1 credit per 36h.

- **Backoff:** Elevated AFR or anomaly spikes reduce CCC regionally.
- **Template quotas:** Caps per mission type to avoid over-farming a single template.

5.8 Mission Templates and Oracles

Template	Evidence Mix	Key Checks	Payout Basis
Rural water pump restore	Geotagged media + NGO receipt + IoT uptime	AOI radius, before/after delta, uptime ≥ policy	Fixed PjP_jPj × uptime tier
School supplies delivery	On-chain attestation + photos + signatures	Route plausibility, beneficiary count	PjP_jPj per beneficiary (capped)
Community mediation	Third-party attest + narrative + timestamp	Duplicate identity/device, time consistency	Flat PjP_jPj

5.9 Settlement Tiers and Disputes

- **Tier-0 Auto-Settle:** Accept/deny; receipt emitted.
- **Tier-1 Review:** Sampled audits; response within policy SLA; partial acceptance possible.
- **Tier-2 Arbitration:** External verifier; binding outcome; receipts reference arbitrator ID.
- **Dispute window:** 72h from decision; new evidence allowed if within schema.

5.10 KPIs and Operating Targets - Dashboard

KPI	Definition	Target / Guard

Precision (audited)	TP / (TP+FP) on Tier-1	≥ 0.95
Recall (audited)	TP / (TP+FN) on Tier-1	Optimize subject to budget
AFR	Audit-Fail Rate	≤ 2% per template
Duplicate Device Rate	Share of claims from reused devices within cooldown	≤ 3%
Geo Drift	Median distance from AOI center	≤ policy by template
Review Latency	Tier-1 median SLA	≤ 72h
Budget Utilization	Payouts / Budget per epoch	85–100% (rollover OK)

5.11 Scenario Modeling and Sensitivities

Adversarial pressure. Suppose a fraction π of submissions originates from coordinated fraud rings. Under current thresholds, expected approved share $A(\pi)$ satisfies:

$$A(\pi) \approx (1-\pi)\cdot \text{TPR} + \pi \cdot (1-\text{TNR})$$

with TPR and TNR measured on sampled audits. Adaptive controls raise $S_{\min}$ or increase Tier-1 rate to keep AFR ≤ 2%.

Throughput regimes.

Regime	Incoming Submissions	Approved Rate	AFR	Notes
Bear	Low–Med	45–65%	≤ 1.5%	Tight sampling; conservative SminS_{\min}Smin
Base	Medium	55–70%	≤ 2.0%	Balanced precision/recall
Hypergrowth	High	60–75%	≤ 2.0%	Expanded audits; throttles mitigate drift

5.12 Privacy and Data Handling

- **Data minimization:**

  Only essential fields stored; sensitive media hashed and prunable.

- **Selective disclosure:**

  Third-party verifiers access least-privilege data via expiring grants.

- **Retention:**

  Evidence TTLs tied to dispute windows; analytics derived features retained in aggregate only.

5.13 Machine-Readable Receipts (Visual)


```json
{
  "action": "mission_claim_decision",
  "template_id": "WATER-RESTORE-v2",
  "identity_hash": "idc_7f…",
  "evidence_refs": ["ipfs://…", "solana:tx/…"],
  "score_S": 0.71,
  "anomaly_A": 0.22,
  "decision": "accepted",
  "payout_pre_burn": 120.00,
  "payout_net": 119.10,
  "burn_rate": 0.0075,
  "audit_tier": "T0",
  "policy_ref": "AnnexA.v1.4",
  "signer": "MPC:PS-MISSIONS-01",
  "timestamp": 1731302400
}
```


5.14 Risks and Mitigations HII/Missions

Risk	Description	Likelihood	Impact	Mitigation	Residual
Evidence Spoofing	Forged media/metadata	Medium	Medium–High	Multi-signal cross-checks; device uniqueness; EXIF/ML forensics	Low–Med

Fraud Rings	Coordinated clusters	Medium	High	Graph/ring detection; ring-fencing; Tier-2 audits; throttles	Low–Med
Oracle/Device Outages	Data gaps	Low–Med	Medium	Graceful degradation; fallback signals; SLA monitoring	Low
Privacy Leakage	Excessive PII retention	Low	High	Data minimization; expiring grants; hashed storage	Low
Reviewer Drift	Inconsistent audits	Medium	Medium	Calibration sets; inter-rater checks; playbooks	Low–Med
Regional Abuse	AOI gaming / geo-spoofing	Medium	Medium	Geo-temporal plausibility; AOI randomization; K-factors	Low–Med

5.15 Visual Index (Will be injected in the final report)

- **Fig. 5-A:** Evidence pipeline (ingestion → validation → scoring → decision).
- **Fig. 5-B:** ROC/PR curves by template with SminS_{\min}Smin.
- **Fig. 5-C:** Sankey of evidence weights into final score SSS.
- **Fig. 5-D:** Heatmap of anomaly score AAA vs. acceptance.
- **Fig. 5-E:** Audit sampling flow (Tier-0/1/2) and latency bands.
- **Fig. 5-F:** Regional geo-drift monitoring with thresholds.

Section 6

Economic Model and Finalization

6.0 Design Principles

- **Stock-to-Flow Predictability:**

  Long-term scarcity maintained through a fixed supply with predictable deflationary pressure generated from action-coupled burns and surplus-driven buybacks.

- **Monetary Stability:**

  Built-in market controls and fail-closed mechanisms minimize discretionary influence, emphasizing data-driven, mechanical scarcity.

- **Impact-Coupled Burns:**

  Supply reduction is linked directly to verified human-impact activity (HII), ensuring that value accrues only with measurable, validated outcomes.

- **Flexibility:**

  Elastic buybacks, minting schedules, and parameter adjustments are bounded by governance and safeguarded against overreach.

6.1 Monetary Stock-Flow Identity

The relationship between the circulating supply (StS_tSt), total supply (SmaxS_{\text{max}}Smax), and market dynamics is governed by a set of equations that tie monetary issuance to verifiable human actions. Over time, the supply is influenced by both the emission rate (MtM_tMt) and the deflationary burn mechanism (BtB_tBt):

St+1=St+Mt−BtS_{t+1} = S_t + M_t - B_tSt+1=St+Mt−Bt

Where:

- M_t is the total minted reward at time t, defined by the mission claims and vesting,
- B_t is the burn rate, tied directly to verified impact ($B_t = \text{Claim-time Burn}(R_t)$),
- S_t is the circulating supply at time t.

Burn dynamics increase over time as the reward system expands, pushing the economy into a deflationary loop once critical thresholds of Clean GMV are met.

6.2 Claim-Time Burn Dynamics

Claim-time burn is tied to HII-verified rewards, scaling with the size of the rewards and the verifiable actions:

$$B^{\text{claim}}_t = 0.0075 \times R_t$$

Where:

- R_t is the reward available to be claimed at time t for each verified mission,
- The burn rate is calculated at the time of claiming and directly reduces the circulating supply.

This burn rate ensures that as missions scale in verified impact, the corresponding reward (and thus supply) is deflated proportionally, maintaining a balance between token inflation and impact.

6.3 Surplus-to-Buyback Controller

Surplus funds are generated when Clean GMV exceeds operational costs and is available for buybacks. The buyback amount (B^{bb}_t) is determined by the surplus available and governed by strict parameters to ensure market stability:

$$B^{\text{bb}}_t = \text{Surplus}(t) \times \min(\text{Market Hygiene Guards})$$

Where:

- Surplus(t) is the available capital after operations,

- **Market Hygiene Guards**: Defined by TWAP, freshness, and deviation from price bands.

Buybacks are executed when conditions permit, and if the market hygiene fails, the buyback is automatically aborted, preserving liquidity stability and preventing negative market pressure.

6.4 Allocation Optimality Checks

The **allocation strategy**—between reward issuance, partner allocations, and liquidity provisioning—should always respect the long-term goals of scarcity and solvency. Each allocation must be tracked in terms of both **circulating supply** and **market absorption** to avoid volatility spikes at major unlocks. The **optimal allocation** balances these two factors while maintaining alignment with the broader economic goals.

Optimal Allocation Formula:

The optimal release amount RoptimalR_{\text{optimal}}Roptimal is determined by the market depth at ±1%, projected burn yield, and the available surplus to balance rewards and market absorption:

Roptimal=min(Absorption Threshold, Surplus Available)R_{\text{optimal}} = \min\left(\text{Absorption Threshold}, \; \text{Surplus Available} \right)Roptimal=min(Absorption Threshold,Surplus Available)

Where:

- **Absorption Threshold** is derived from DEX depth, trading volume, and volatility parameters.
- **Surplus Available** is the portion of operating surplus that is eligible for buybacks or reward funding, based on governance decisions and market health.

6.5 Liquidity Microstructure Model

Liquidity provisioning is fundamental to the stability and market depth of $PEACE. The liquidity pool (LP) is engineered to absorb routine flow and maintain tight realized spreads. The **depth** and **slippage** for the token are defined by the liquidity band structure, with primary and secondary bands adjusting based on real-time market dynamics.

Liquidity Targeting:

- **Core Depth:** Set to absorb 55-65% of regular trades within the ±1% price range.
- **Near Bands:** Capture 25–35% of trades within the ±1%–±2% price range.
- **Guard Bands:** Capture 5–15% beyond ±2%, ensuring stability during high volatility or large trades.

Liquidity Depth Formula:

The liquidity depth D±wD_{\pm w}D±w at a given price range www is a function of trading volume, market volatility, and order book density. For a price change Δp\Delta pΔp, the realized slippage s(Δp)s(\Delta p)s(Δp) is calculated as:

s(Δp)=xD±ws(\Delta p) = \frac{x}{D_{\pm w}}s(Δp)=D±wx

Where:

- xxx is the notional trade size,
- D±wD_{\pm w}D±w is the available depth at price range www from the market price.

Liquidity Sensitivity Model:
For large trades, s(Δp)s(\Delta p)s(Δp) increases as D±wD_{\pm w}D±w decreases, creating an inverse relationship between available liquidity and market volatility. This relationship helps set thresholds for when buybacks should be paused or liquidity bands adjusted.

6.6 Governance Capture Resistance

The tokenomics design has built-in mechanisms to prevent governance capture:

- **Identity-Capped Voting:** Each identity can vote only up to 0.1% of the total token supply, ensuring decentralization of decision-making.
- **Classed Quorums and Timelocks:** Changes to the protocol, especially those impacting monetary policy or governance structure, require a **supermajority quorum** and are subject to timelocks, ensuring sufficient review time and accountability.
- **Class-3 Proposals:** Changes to constitutional aspects (supply policy, identity cap, etc.) are governed by a higher quorum (≥60%) and an extended timelock (96h minimum).

6.7 Reconciliation and Auditability

Reconciliation refers to the process of verifying the total supply and the actions taken within the system against the published schedules, emissions, and buyback actions. It is critical for ensuring that the protocol operates transparently and in line with governance decisions.

All key parameters, including supply, burn rates, and rewards, must be auditable via machine-readable receipts. The **Parameter Registry (Annex A)** includes both the current values and historical snapshots of all relevant parameters, ensuring that any changes are publicly available for verification by stakeholders.

6.8 KPIs, Alerts, and Control Mapping

KPI	Definition	Target	Alert on Breach
Supply Growth Rate	Annualized increase in circulating supply	≤ 5%	If >5%, review emission cadence
Buyback Frequency	Monthly execution of buybacks	Monthly target	No execution due to hygiene failure
DEX Depth ±1%	Available depth at ±1% price	≥ 90% of target	If below target, pause rewards
Burn Yield	Claim-time burn per epoch	≥ 0.75%	If yield <0.75%, check reward scaling
Liquidity Utilization	Depth absorption percentage	80–90%	If outside range, readjust liquidity bands
Runway Stability (CM)	Minimum CM floor	≥18 months	If below floor, pause non-essential operations

6.9 Scenario Stress Tests

- **Base Case:**

  20% annual GMV growth, stable market conditions, 1.5% burn yield.

- **Bear Case:**

  5% GMV growth, market volatility spikes, buybacks inactive for 6 months.

- **Hypergrowth Case:**

  50% GMV growth, full burn yield, liquidity depth expansion.

Projected Supply Reduction:

Scenario	Annual Supply Reduction	Buyback Frequency	Resulting Depth Change
Base	~2.5%	3–4 times per month	Stable market depth, regular buybacks
Bear	~1.0%	0–2 times per month	Shallow liquidity, depth testing
Hypergrowth	~5.0%	5–7 times per month	Increased market depth, high-frequency buybacks

6.10 Risks and Mitigations-Economic Model-Specific

Risk	Description	Likelihood	Impact	Mitigation	Residual
Supply Inflation	Emissions exceed targeted deflation	Low–Medium	High	Adjust emission rates; slow down token issuance	Low
Liquidity Drain	LP depth unable to absorb large trades	Medium	High	Dynamic LP rebalancing; mirroring pools	Low–Medium
Buyback Failures	Market conditions prevent buyback execution	Medium	Medium	Hygiene checks; threshold triggers for pausing buybacks	Low
Market Volatility	Price slippage due to large market movements	Medium	High	Adaptive liquidity; deferred buybacks during high volatility	Low

6.11 Long-Term Equilibrium and Sustainability

The long-term economic model targets a **Stock-to-Flow** ratio that provides a sustainable, **2–5% annual deflationary pressure**. This is achieved by linking burn rates directly to verified economic activity, with surplus-only buybacks adjusting to market conditions.

Simulated Deflationary Pressure:

- **Base case (20% GMV growth):**

  Projected annual supply reduction of ~2–3%.

- **Hypergrowth case (50% GMV growth):**

  Projected annual supply reduction of ~5–7%.

- **Sustainability Indicator:**

  Stock-to-Flow ratio in alignment with similar models (e.g., Uniswap's fee switch) targets a stable price floor that can weather market cycles.

Section 7

Peace Store Economy and Merchant Dynamics

7.0 Design Principles

- **Impact-Coupled Commerce:**

Store throughput (Clean GMV) anchors monetary sinks and surplus formation, completing the loop from verified missions to durable scarcity.

- **Merchant-First Reliability:**

  Deterministic settlement, transparent fees, and predictable discount policy drive adoption and retention.

- **Defense-in-Depth:**

  Fraud controls, dispute tiers, and compliance modes reduce operational and regulatory risk.

- **Receipts Everywhere:**

  Machine-verifiable receipts enable third-party recomputation of fees, discounts, surplus, and payouts.

7.1 Quick Parameter Reference (Annex A)

Surface	Parameter	Policy / Range	Default	Notes
Payment Currencies	Accepted media	$PEACE; selected stables	Policy set	$PEACE prioritized via discounts
Store Fee (Take Rate)	% of order notional	0.8–3.0%	**1.5%**	Dynamic within bounds
$PEACE Discount Cap	% price reduction	0–30%	**≤30%**	Budgeted and audited
Settlement Cadence	Merchant payout	T+1…T+7	**T+2**	Varies by risk tier
Refund Window	Buyer-initiated	7–30 days	**14 days**	Policy & category specific
Chargeback Reserve	% held in reserve	0–5%	**2%**	Risk-weighted

Evidence SLA	Dispute evidence submission	24–96h	**72h**	Per dispute tier
Surplus Routing	Fee surplus → treasury	Rules-based	Enabled	Eligibility under CM floors

7.2 Merchant Onboarding and Eligibility

- **Identity and Risk Tiering:** Merchants undergo identity verification and risk scoring (Tier-A/B/C) affecting fee tier, settlement cadence, and reserve ratio.
- **Catalog Controls:** Each SKU includes category, refund policy, evidence requirements (e.g., delivery confirmation schema), and region flags.
- **Compliance Modes:** Open or KYC modes applied per jurisdiction; mode changes governed constitutionally (see Section 4.8).

Activation Formula (visual). Merchant activation eligibility requires:

scoreonboard ≥ θmin∧policy_docs=valid∧risk_tier∈{A,B,C}\text{score}_{\text{onboard}} \;\ge\; \theta_{\text{min}} \quad\wedge\quad \text{policy_docs} = \text{valid} \quad\wedge\quad \text{risk_tier} \in \{\text{A,B,C}\}scoreonboard≥θmin∧policy_docs=valid∧risk_tier∈{A,B,C}

7.3 Pricing, Fees, and Discount Policy

- **Base Price PPP:** Merchant-set.
- **Policy Fee fff:** Store take rate within bounds.
- **$PEACE Discount ddd:** Bounded by discount cap and budget.
- **Net Price to Buyer:**

Pnet=P⋅(1−d)P_{\text{net}} = P \cdot (1-d)Pnet=P⋅(1−d)

- **Merchant Payout (pre-reserve):**

Πgross=P⋅(1−d)⋅(1−f)\Pi_{\text{gross}} = P \cdot (1-d) \cdot (1-f)Πgross=P⋅(1−d)⋅(1−f)

- **Reserve Holdback:** Π=Πgross⋅(1−r)\Pi = \Pi_{\text{gross}} \cdot (1-r)Π=Πgross⋅(1−r), where rrr is risk-weighted reserve (released after the refund/chargeback window).

Discount Budgeting.

A per-epoch discount budget BdB_dBd prevents over-subsidization:

∑ordersP⋅d ≤ Bd\sum_{\text{orders}} P \cdot d \;\le\; B_dorders∑P⋅d≤Bd

Budget usage appears in monthly manifests; overruns are mechanically blocked.

7.4 Payment and Settlement Flows

1. **Checkout:** Buyer selects currency; discount logic applies if $PEACE is used and budget permits.
2. **Escrow:** Funds enter an escrow program awaiting fulfillment evidence.
3. **Fulfillment:** Merchant submits evidence per SKU policy (e.g., delivery attestation).
4. **Settlement:** Escrow releases to merchant per cadence; reserves held as configured.
5. **Accounting:** Fees, reserves, and discount usage posted to receipts; surplus routed to treasury under policy.

Clean GMV.

Clean GMV=Gross GMV−Refunds−Chargebacks−Fraud_write_offs\text{Clean GMV} = \text{Gross GMV} - \text{Refunds} - \text{Chargebacks} - \text{Fraud_write_offs}Clean GMV=Gross GMV−Refunds−Chargebacks−Fraud_write_offs

7.5 Unit Economics and Growth Levers

- **Take Rate (TR):** Policy-bounded; sensitive to category and risk tier.
- **Contribution Margin (Store):**

CMstore=TR⋅Clean GMV−Ops Cost−Discount Spend\text{CM}_{\text{store}} = \text{TR} \cdot \text{Clean GMV} - \text{Ops Cost} - \text{Discount Spend}CMstore=TR⋅Clean GMV−Ops Cost−Discount Spend

- **Merchant LTV (Visual):**

LTVm=∑t=1T(GMVm,t⋅TR−csupport,t−crisk,t)⋅1(1+ρ)t\text{LTV}_{m} = \sum_{t=1}^{T} \Big(\text{GMV}_{m,t}\cdot \text{TR} - c_{\text{support},t} - c_{\text{risk},t}\Big)\cdot \frac{1}{(1+\rho)^t}LTVm=t=1∑T(GMVm,t⋅TR−csupport,t−crisk,t)⋅(1+ρ)t1

- **CAC and Payback:** CAC measured per acquisition channel; payback === CAC / (monthly contribution).
- **Growth Levers:** Discount targeting (within budget), SKU curation, partner bundles, and mission-linked campaigns.

7.6 Marketplace Dynamics and Reputation

- **Matching and Ranking:** Search ranks by relevance, fulfillment success rate, dispute history, and $PEACE acceptance.
- **Reputation Score:** Rolling composite of on-time fulfillments, dispute outcomes, and refund rate; decays over time to reward consistent performance.
- **Dispute Tiers:**
    - *Tier-0:* Auto-resolve with evidence (e.g., delivery proof).
    - *Tier-1:* Human review within SLA.
    - *Tier-2:* External arbitration; binding outcome.

7.7 Risk Management, Compliance, and Fraud Controls

- **Fraud Analytics:** Device/wallet clustering, abnormal refund patterns, SKU outliers, velocity limits.

- **Chargebacks:** Reserves released after window expiry; elevated rates automatically increase rrr and may slow cadence.
- **Volatility Handling:** $PEACE payments may convert to stables at execution price under policy; conversions generate receipts.
- **Compliance:** Mode flags (Open/KYC) enforced at payment gateway; region-specific restrictions apply per SKU.

Incident Guardrails. If refund or chargeback rates breach thresholds, automatic actions include fee adjustments within bounds, reserve increases, or temporary discount reduction.

7.8 KPIs and Operating Targets-Dashboard

KPI	Definition	Target / Guard
Clean GMV	GMV net of refunds/chargebacks/fraud	Upward trend; seasonally adjusted
Take Rate (TR)	Average fee rate	Within policy band; stable
Discount Spend Ratio	Discount spend / Gross GMV	≤ budget; trending efficient
Refund Rate	Refunds / Orders	≤ policy by category
Chargeback Rate	Chargebacks / Orders	≤ policy; declining
Merchant Retention (90d)	% of active merchants retained	≥ target cohort percentile
Order Approval Latency	Time to auto-approval or review	≤ policy SLA
Surplus Yield	Surplus / Clean GMV	≥ target; contributes to buyback eligibility

7.9 Scenario Modeling and Sensitivities

Discount Sensitivity (Visual).

ddd (avg. discount)	Conversion Lift	Discount Spend Ratio	Net CM Impact
5%	+3–6%	1.0–1.5%	Neutral–Positive
10%	+7–12%	2.0–3.0%	Positive if lift ≥ 8%
20%	+12–22%	4.0–6.0%	Positive only with strong lift and low refunds
30%	+18–32%	6.0–9.0%	Use selectively; budget-capped

Refund/Chargeback Stress.

Shock	Parameter Shift	Automatic Response
Refund spike	Refund rate +300 bps	Tighten evidence; raise reserve rrr; slow settlements to T+5
Chargeback spike	+150 bps sustained	Increase rrr; elevate review tier; lower discount cap temporarily
Volatility shock	Price drift beyond fence	Convert $PEACE leg at execution; pause non-essential discounts

7.10 Machine-Readable Receipts

Order Settlement


```
{
  "action": "order_settle",
  "order_id": "ORD-7QJ2",
  "buyer": "addr_b_…",
  "merchant": "addr_m_…",
  "notional": 150.00,
  "currency": "PEACE",
  "discount": 0.10,
  "fee": 0.015,
  "reserve": 0.02,
  "payout": 150*(1-0.10)*(1-0.015)*(1-0.02),
  "tx_hash": "8k…Rs",
```

```
  "policy_ref": "AnnexA.v1.4",
  "timestamp": 1731309600
}
```

Surplus Accrual

```
{
  "action": "store_surplus_post",
  "epoch": 312,
  "clean_gmv": 285000.00,
  "fees_collected": 4275.00,
  "discount_spend": 3800.00,
  "ops_cost": 900.00,
  "surplus": -425.00,
  "eligibility": "ineligible (CM floor / negative)",
  "policy_ref": "AnnexA.v1.4"
}
```

Currency Conversion (if applicable)

```
{
  "action": "payment_convert",
  "order_id": "ORD-7QJ2",
  "from": "PEACE",
  "to": "USDC",
  "rate": 0.97,
  "twap_window_s": 7200,
  "freshness_s": 42,
  "deviation_ok": true,
  "tx_hash": "3P…9X"
}
```

7.11 Risks and Mitigations Store/Merchant

Risk	Description	Likelihood	Impact	Mitigation	Residual
Refund / Chargeback Surge	Elevated post-sale reversals	Medium	Medium–High	Reserves, evidence tightening, cadence slowdown	Low–Med
Fraud Rings	Coordinated buyer/merchant schemes	Medium	High	Graph analytics; velocity limits; ring-fencing; Tier-2 reviews	Low–Med
Discount Overspend	Budget breach or misallocation	Low–Med	Medium	Hard budget caps; real-time throttles	Low
Liquidity/Volatility Spillover	Price instability from large $PEACE legs	Medium	Medium	Conversions at execution; deviation fences	Low–Med
Compliance Event	Jurisdictional rule change	Low–Med	High	Mode switches; SKU geofencing; receipts for audit	Med
Reputation Shocks	Merchant failures or quality drops	Medium	Medium	Ranking penalties; reserve hikes; targeted outreach	Low–Med

---

7.12 Visual Index (will be injected in to the final report)

- **Fig. 7-A:** Checkout and settlement flow (escrow → evidence → payout → receipts).
- **Fig. 7-B:** Take-rate and discount waterfalls (buyer price → merchant payout → surplus).
- **Fig. 7-C:** Clean GMV vs. Discount Spend and Surplus contribution (time series).
- **Fig. 7-D:** Refund/chargeback stress with automatic policy responses.
- **Fig. 7-E:** Merchant reputation dashboard (fulfillment, disputes, retention).

Section 8

Identity, Passport, and HII-Linked Governance

8.0 Design Principles

- **Uniqueness before scale:**

  Governance legitimacy requires one-person-one-cap mechanics with proof of uniqueness as the foundation.

- **Privacy by construction:**

  Selective disclosure and minimal data retention preserve safety without degrading integrity.

- **Receipted authority:**

  Every identity event (issue, re-verify, suspend, revoke) emits verifiable receipts.

- **Mode flexibility:**

  Open and KYC modes coexist behind a single gateway, switchable by constitutional governance.

8.1 Quick Parameter Reference (Annex A)

Surface	Parameter	Policy / Range	Default	Notes
Identity Cap	Max counted voting weight per identity	0.05–0.20% of supply	**0.10%**	Applied after delegation aggregation

Credential TTL	Passport credential validity	90–365 days	**180 days**	Requires periodic re-verification
Liveness Method	Primary liveness modality	Multi-factor	**Active challenge**	With anti-replay tokens
Device Cooldown	Reuse delay per device fingerprint	1–30 days	**7 days**	Reduces farmed identities
Sybil Score Gate	Max allowed Sybil risk score	0.0–1.0	**≤ 0.35**	Above gate → manual/advanced proof
KYC Mode	Jurisdictional mode flag	Open / KYC	**Open**	Constitutional to change
Data Retention	Evidence TTL	7–365 days	**90 days**	Post-dispute purge; hashed archives
Receipt Requirement	Identity events emitting receipts	Boolean	**Enabled**	Hashes and policy references

8.2 Identity Model and Credential Stack

- **DID & VC stack:**

  Each participant holds a decentralized identifier (DID) and one or more verifiable credentials (VCs) attesting to uniqueness, regional eligibility, and optional KYC status.

- **Passport (non-transferable):**

A non-transferable credential (“Passport”) bound to the DID serves as the governance and program access key. Transfer attempts or credential misuse trigger suspension.

- **Tiering:**
  - Tier-O Open: Pseudonymous uniqueness proofs; suitable for governance with capped weight and for low-risk missions.
  - Tier-K KYC: Full KYC credential for restricted jurisdictions or roles (e.g., large merchant payouts).
  - Tier-A Attested Entity: NGO/merchant credentials with additional attestations (legal entity, banked status).

8.3 Enrollment and Verification Flows

Enrollment pipeline

1. **DID creation** → 2. **Active liveness challenge** (nonce-bound, anti-replay) → 3. **Device fingerprint** bind with cooldown → 4. **Uniqueness heuristics** (graph and clustering) → 5. **Optional KYC** per jurisdiction → 6. **Passport issuance** (VC + on-chain reference).

Re-verification

- Triggered by TTL expiry, risk score elevation, or suspicious cluster linkages. Re-verification must pass current thresholds; otherwise the passport is suspended with a dispute window.

Suspension & Revocation

- Suspensions occur automatically on significant risk score spikes; revocations require adjudication or failure to re-verify within SLA. All state changes emit receipts.

8.4 Privacy and Data Minimization

- **Selective disclosure:**

  Zero-knowledge or selective-field VCs expose only required attributes (e.g., “over-18,” “region=EU”) without revealing PII.

- **Ephemeral evidence:**

  Sensitive liveness media stored off-chain with short TTL; on-chain references contain only hashes, timestamps, and policy pointers.

- **Access control:**

  Verifiers receive least-privilege access with time-boxed grants; all access is logged and hashed for audit.

8.5 Governance Integration

Voting cap.

With total supply SSS and identity cap rate ccc (e.g., 0.1%), maximum counted weight per identity is:

wmax=c·Sw_{\max} = c \cdot Swmax=c⋅S

For identity jjj with stake sjs_jsj and inbound delegations djd_jdj:

wj=min(sj+dj, wmax)w_j = \min\big(s_j + d_j,\; w_{\max}\big)wj=min(sj+dj,wmax)

Delegation rules

- Delegations auto-expire after the policy interval; renewal requires fresh signature.
- Delegation graphs are acyclic; inbound delegate count may be bounded (policy surface).
- Revoked or suspended passports invalidate delegated weight until restoration.

Proposal gating

- Proposal creation rights require a valid passport and minimum endorsements from distinct identities or a refundable deposit, preventing spam.

8.6 Abuse Resistance and Sybil Controls

- **Clustering analytics:** Graphs of device, network, timing, and behavioral features identify rings; edges above risk thresholds trigger ring-fencing.
- **Velocity limits:** Per-epoch caps on new passports per device/IP ASN; bursts auto-throttle.
- **Challenge diversity:** Periodic rotation of liveness challenges and sensor mixes reduces replay risk.
- **Back-pressure:** Elevated regional Sybil risk tightens gates (higher liveness strictness, longer cooldowns).
- **Escalation path:** Identities failing heuristics can pass higher-assurance proofs (e.g., Tier-K KYC) to proceed.

8.7 HII Linkage and Mission Binding

- **Claim binding:**

  Each mission claim references a passport hash; claims from suspended or revoked passports are rejected.

- **Per-template uniqueness:**

  Templates may require proof of local presence (AOI radius), time-bounded liveness, or NGO co-attestation.

- **Rate limits:**

  Identity-level credits (Section 5.7) prevent rapid multi-claim patterns even when uniqueness holds.

8.8 Compliance Modes and Jurisdiction Handling

- **Mode switch:**

Compliance mode (Open/KYC) is a constitutional parameter. Mode changes propagate to gateway checks and Passport issuance rules.

- **Geo-gating:**

  Region attributes in VCs gate participation in sales, missions, and merchant payouts according to policy.

- **Records:**

  Compliance actions emit receipts (mode, basis, jurisdiction code) to support audit trails without publishing PII.

8.9 Machine-Readable Receipts (Visual)

Passport issue


```
{
  "action": "passport_issue",
  "did": "did:ps:8f…",
  "vc_id": "vc_2e…",
  "tier": "Tier-0",
  "ttl_days": 180,
  "risk_score": 0.12,
  "policy_ref": "AnnexA.v1.4",
  "timestamp": 1731313200
}
```


Re-verify / (suspend / revoke)


```
{
  "action": "passport_update",
```

```
  "did": "did:ps:8f…",
  "event": "suspend",
  "reason_code": "sybil_cluster_link",
  "risk_score": 0.51,
  "effective_until": 1731918000,
  "dispute_window_h": 72,
  "policy_ref": "AnnexA.v1.4
```

Governance vote (identity-capped)

```
{
  "action": "gov_vote",
  "proposal_id": "C2-2025-045",
  "did": "did:ps:8f…",
  "weight_applied": 0.001*S,
  "cap_rate": 0.001,
  "delegate_of": null,
  "timestamp": 1731403200
}
```

8.10 KPIs and Operating Targets-Dashboard

KPI	Definition	Target / Guard
Verified Identities	Active passports within TTL	Upward trend; balanced across regions
Re-verification Coverage	% passports ≤ 7 days to expiry successfully renewed	≥ 95%
Sybil Fail Rate	Share of applications failing Sybil gate	Stable; ≤ policy band with low false positives
False Positive Rate	% appeals that overturn suspensions	≤ 2%
Liveness Pass Rate	Passes / total challenges	≥ 98% (Open); ≥ 99% (KYC)
Delegation Churn	% delegations expiring without renewal per epoch	Within normal band; downward trend
Compliance Exceptions	Mode-related denials	Near-zero; justified by receipts
Governance Participation	Unique identities per epoch	At or above class quorum expectations

8.11 Risks and Mitigations - Identity/Passport

Risk	Description	Likelihood	Impact	Mitigation	Resid ual
Sybil Clusters	Coordinated farms bypass heuristics	Medium	High	Multi-signal gating; escalation to Tier-K; device cooldowns; ring-fencing	Low–Med
Issuer Centralization	Over-reliance on a single verifier	Low–Med	High	Multi-issuer VCs; quorum-based attestation sets; fallback vendors	Low
Liveness Bypass	Deepfake/replay attacks	Medium	Mediu m–Hig h	Active challenges; hardware-bound tokens; rotating modalities	Low–Med
Privacy Leakage	Mishandled evidence	Low	High	Selective disclosure; short TTL; hashed logs; access receipts	Low
Regional Compliance Shift	Sudden KYC mandates	Medium	High	Constitutional mode switch; geo-gating; accelerated Tier-K migration	Med
Denial of Service	Enrollment floods	Medium	Mediu m	Velocity limits; proof-of-work or deposit gates; queueing	Low–Med

8.12 Visual Index (Will be added in to the Final report)

- **Fig. 8-A:** Identity gateway flow (enroll → verify → issue → re-verify → revoke).
- **Fig. 8-B:** DID/VC stack with Passport binding and selective disclosure.
- **Fig. 8-C:** Governance cap application (stake + delegations → min with cap).
- **Fig. 8-D:** Sybil detection graph and ring-fencing workflow.
- **Fig. 8-E:** Compliance mode switch sequence and effect propagation.

Section 9

Security, Audits, and Incident Response

9.0 Security Model and Principles

- **Least Privilege and Separation of Duties:**

  Privileged actions are partitioned across roles, keys, and systems; no single operator can execute end-to-end sensitive flows.

- **Defense in Depth:**

  Layered controls across smart contracts, key custody, infrastructure, and processes; failures degrade safely.

- **Fail-Closed Execution:**

  Monetary and governance surfaces prefer automatic abort over uncertain behavior.

- **Attestable Operations:**

  All privileged events produce signed, machine-verifiable receipts with policy references and payload hashes.

- **Continuous Assurance:**

  Proactive monitoring, recurring audits, and reproducible builds maintain integrity between releases.

9.1 Quick Parameter Reference (Annex A)

Surface	Parameter	Policy / Range	Default
Key Custody	MPC quorum	2/3…4/7	**3/5**
Custody Devices	HSM/TEE requirement	Required for signers	**MPC + HSM**
Rotation	Key rotation cadence	60–180 days	**90 days**
Release	Build attestation level	SLSA-3…SLSA-4	**SLSA-3+**
Release	SBOM generation	Every release	**Enabled**
Contracts	Upgradability policy	Proxy with guarded gateway or immutable	**Gateway-guarded**
Monitoring	Oracle freshness cap	10–180s	**≤120s**
Incident	Emergency pause TTL	6–48h	**24h**
Bounty	Max payout tier	Severity-based	**Critical tier enabled**

9.2 Threat Model

Assets:

Treasury funds, LP positions, token contracts, governance gateway, identity/passport data, marketplaces, build/signing pipelines.
Adversaries:

External exploiters, collusive rings, malicious insiders below quorum, supply-chain actors, data scrapers, advanced MEV bots.
Surfaces:

Smart contracts, MPC endpoints, oracles, AMM rebalances, governance payloads, API gateways, CI/CD, artifact registries, vendor integrations.

9.3 Key Management and Custody

- **MPC + HSM:** Treasury, governance, and deployer keys require multi-party computation with hardware-backed signers. No hot single-key control.

- **Quorum Design:** **3/5** for routine treasury ops; **4/7** for constitutional payloads. Emergency pause requires a distinct quorum and expires automatically.

- **Rotation and Recovery:** Scheduled rotation every **90 days**; recovery ceremonies documented, tested on a staging fork, and attested.

- **Access Hygiene:** Role-scoped signing rights, per-signer velocity caps, time-boxed approvals, and geo/behavioral anomaly detection.

9.4 Build, Supply Chain, and Release Integrity

- **Reproducible Builds:** Deterministic builds for contracts and off-chain controllers; sha256 digests published.

- **SLSA and SBOM:** Build attestations (SLSA-3+) and per-release SBOMs; dependency diffs gated by policy.

- **Code Signing:** Artifacts signed with dedicated release keys (separate from treasury/governance).

- **Release Gate:** No deployment without (i) passing invariant tests, (ii) updated SBOM, (iii) unchanged policy diffs or approved C2/C3 governance.

9.5 Smart Contract Assurance

- **Formal Reviews:** Pre-TGE audits covering token, vesting, escrow, governance gateway, and liquidity controllers.

- **Verification and Fuzzing:** Property-based tests, invariant suites (pause behavior, supply conservation, bounds enforcement), and continuous fuzzing on staging.

- **Upgrade Policy:** Where proxies exist, implementations can change only via governance receipts; immutable modules are preferred where feasible.
- **Invariants (illustrative):**
    - Supply never exceeds fixed max.
    - Buybacks cannot execute without hygiene pass.
    - Unlocks cannot exceed calendar and absorption caps.
    - Governance payloads must hash-match proposal receipts.

9.6 Infrastructure Security

- **Network Segmentation:** Isolated signer subnets; no inbound from public internet; bastion-only egress.
- **Secrets:** Hardware-backed KMS, short-lived tokens, and sealed environment variables; no plaintext secrets in CI.
- **Access Control:** Just-in-time access, MFA, and peer approval; session recording on privileged hops.
- **DoS and Abuse:** Rate limits, WAF/CDN shielding, and circuit breakers for API spikes.
- **Logging and SIEM:** Immutable logs, time-sync via secure NTP, and centralized SIEM with alerting SLAs.

9.7 Data Security and Privacy

- **Minimization:** Only essential fields retained; PII segregated and tokenized.
- **Encryption:** TLS 1.3 in transit; AES-256 at rest; per-tenant keys where applicable.
- **TTL and Purge:** Evidence TTL per policy; hashed archives for audit only.
- **Access Grants:** Time-boxed, least-privilege access to reviewers; full audit trail with hashed receipts.

9.8 Monitoring and Observability

- **On-Chain:** Oracle freshness, deviation fences, failed governance payloads, abnormal unlocks, LP position health.

- **Off-Chain:** API latency/error spikes, signer availability, CI integrity, anomaly scores on identity influx.
- **Alert SLA:** P1 page in ≤5 minutes; escalation to on-call quorum; automated freeze on certain signatures (e.g., oracle freshness breach).

Operational Metrics (visual)

Metric	Guard / Target
Oracle freshness fail rate	≤ 0.1% of epochs
Controller aborts (buyback)	Healthy baseline; investigate bursts
Privileged action latency	Within class timelock windows
Signer availability	≥ 99.9%
CI attestation coverage	100% of releases

9.9 Incident Response and Business Continuity

Objectives: Contain blast radius, preserve solvency, protect users, and restore normal operations with documented RCAs.

9.9.1 Severity Matrix and Objectives

Sev	Description	Examples	Target RTO	Target RPO
S0	Catastrophic	Treasury compromise, critical contract invariant break	1h	0–5m
S1	High	Oracle outage, governance gateway malfunction	4h	≤15m
S2	Medium	Regional service degradation, API abuse	24h	≤1h
S3	Low	Non-critical bugs, reporting errors	72h	≤24h

9.9.2 Runbooks-abridged

- **Contract/Protocol Threat (S0):** Trigger emergency pause (time-boxed), revoke/rotate affected keys, snapshot state, publish incident receipt, assemble patch via governance or redeploy immutable fallback.
- **Oracle Freshness/Deviation Failure (S1):** Freeze buybacks and unlocks, switch to backup oracle, widen bands if needed, issue status bulletin.
- **Liquidity Shock (S1/S2):** Reinforce near bands, suspend discretionary releases, raise monitoring frequency, publish market quality telemetry.
- **Data Exposure (S1):** Isolate systems, invalidate credentials, notify affected parties under policy, independent forensics, purge and re-key.

9.9.3 Communications

- **Channels:** Status page, signed chain message, and repository advisory.
- **Cadence:** Initial notice ≤1h for S0/S1; updates every 4–8h until resolved; RCA within 7 days.

9.10 Bug Bounty Program

- **Scope:** Core contracts, governance gateway, vesting/escrow, identity middleware, and payment/settlement flows.
- **Safe Harbor:** Good-faith research protected; non-exfiltration and coordinated disclosure required.
- **Tiers (illustrative):**
  - Critical (RCE/funds at risk): up to **$100k**
  - High (loss of funds via bounded vectors): up to **$50k**
  - Medium (stability/DoS): up to **$15k**
  - Low (information disclosure/minor): up to **$5k**
- **SLA:** Triage ≤72h; bounty decision ≤14d post-validation.

9.11 Audit Plan and Cadence

- **Pre-TGE:** Two independent firms; coverage of all production contracts, parameter bounds, and gateway logic; remediation and re-audit required.
- **Post-TGE:** Quarterly light audits on diffs; annual full audit; targeted reviews for major changes (C3).
- **Third-Party Assessments:** Annual penetration tests on APIs and custody systems; red-team exercises for signer workflows.
- **Evidence Pack:** Reports, re-audit confirmations, SBOMs, and build attestations referenced in Annexes.

9.12 Control KPIs and Thresholds

KPI	Definition	Target / Guard	Action on Breach
Invariant Violations	Failing invariant tests in CI or on-chain monitors	0	Freeze release; escalate
Emergency Pauses	# per quarter	≤ 1 (exercises excluded)	Post-mortem; control hardening
Audit Findings	Open items by severity	0 Critical/High	Block release until resolved
Key Rotations	On-time rotation rate	100%	Emergency ceremony if overdue
Attestation Coverage	Releases with valid attestations	100%	Rollback and re-issue

9.13 Risks and Mitigations-Security-Specific

Risk	Description	Likelihood	Impact	Mitigation	Residual
Private Key Compromise	Signer device breach	Low	High	MPC + HSM, quorum, geo/behavioral checks, rapid rotation	Low

Oracle Manipulation	Stale/biased feeds	Low–Med	High	Freshness/deviation fences, multiple sources, fail-closed	Low–Med
Contract Logic Flaw	Undiscovered vuln	Low–Med	High	Audits, fuzzing, invariants, limited upgradability	Low
CI/CD Supply-Chain Attack	Malicious dependency/runner	Low–Med	High	SBOMs, attestations, pinned builders, isolated runners	Low–Med
Governance Exploit	Payload smuggling/capture	Low–Med	High	Gateway schema checks, payload hash, identity cap, staging	Low
Data Exposure	PII or secret leakage	Low	High	Minimization, TTL, encryption, access receipts	Low

9.14 Machine-Readable Receipts (Visual)

Emergency Pause


```
{
  "action": "emergency_pause",
  "reason": "oracle_freshness_breach",
  "ttl_h": 24,
  "policy_ref": "AnnexA.v1.4",
  "signers": ["MPC:PS-SEC-01","MPC:PS-SEC-03","MPC:PS-SEC-05"],
  "tx_hash": "5F…kQ",
  "timestamp": 1731421200
}
```


Key Rotation


```
{
  "action": "key_rotate",
  "keyset": "TREASURY-V2",
  "old_fingerprint": "ab12…",
```

```
  "new_fingerprint": "f3c9…",
  "quorum": "3/5",
  "effective_block": 21998011,
  "attestation": "sha256:…",
  "policy_ref": "AnnexA.v1.4"
}
```


Build Attestation


```
{
  "action": "build_attest",
  "repo": "contracts/vesting",
  "commit": "b7e9c…",
  "slsa_level": "3+",
  "sbom_digest": "sha256:…",
  "artifact_hash": "sha256:…",
  "timestamp": 1731424800
}
```

9.15 Visual Index (will be injected into the final report)

- **Fig. 9-A:** Key custody topology (MPC/HSM, quorums, rotation).
- **Fig. 9-B:** Release pipeline (code → attest → audit → gateway → deploy).
- **Fig. 9-C:** Monitoring map (on-chain/off-chain signals and alerting).
- **Fig. 9-D:** Incident runbook swimlane (S0/S1 timelines).

Section 10

Roadmap, Metrics, and Go-Live Readiness

10.0 Execution Principles

- **Gate-based delivery:**

  Progress advances only when objective gates pass (audit sign-off, liquidity hygiene, governance approvals).

- **Market-quality first:**

  Early phases prioritize spread, depth, and solvency over growth.

- **Receipts and reproducibility:**

  Every gate, change, and deployment emits verifiable receipts; builds are attestable.

- **Minimal variance:**

  Frozen parameters during freeze windows; deviations require elevated governance.

10.1 Phase Roadmap

Phase	Window	Objectives	Primary Gates	Outputs
P0 — Foundations	Pre-TGE (T-90 → T-45)	Contract finalization; audit rounds; parameter registry v1	Dual audit “clean” or remediated; invariant suite green; SBOM + attestations published	Audit reports; Annex A v1; deployment plan
P1 — Launch Prep	T-45 → T-7	Auction configuration; identity gateway; custody/MPC ceremonies	Governance C2 approvals; KYC mode tables where required; address registry sealed	Sale doc; addresses; policy hashes; dry-run receipts

P2 — Discovery & TGE	T-7 → T+3	Run auction; settle p*p^*p*; seed and lock primary LP; mirrors live	Uniform-price settlement receipts; LP lock tx; router checks	p*p^*p*; C0C_0C0; LP band map; mirror pool IDs
P3 — Early Stabilization	T+4 → T+21	Spread/depth stabilization; first KPI pack; incident runbooks primed	Market-quality KPIs ≥ floors for 14 consecutive days	D+7 & D+21 reports; telemetry receipts
P4 — Consolidation	T+22 → T+90	Normalize band density; verify unlock calendar mechanics; first governance cycle	No severity-1 incidents; CM ≥ floor; governance latency within SLA	D+90 report; parameter diffs; ORR sign-off

10.2 Go/No-Go Gates and Objective Criteria

Gate	Metric	Threshold	Evidence
G1 — Code & Audit	Critical/High findings	**0 open**	Auditor letters; re-audit diffs
	Invariant failures	**0**	CI receipts; test logs
G2 — Custody	MPC quorum live	**3/5** (routine), **4/7** (constitutional)	Ceremony receipts
G3 — Governance	Class approvals	**C2/C3** as required	Proposal + exec receipts
G4 — Identity/Compliance	Mode tables	Published, tested	Gateway receipts
G5 — Auction Readiness	Dry-run success	No aborts	Dry-run receipts
G6 — Market Quality	Depth ±1%	≥ policy floor for dress rehearsal	Venue telemetry
G7 — Observability	P1 alerting SLA	≤ 5 min	Pager test receipts

Go/No-Go meeting signs a single **LRR** (Launch Readiness Review) receipt referencing all gate artifacts.

10.3 Metrics and Targets Pre- and Post-TGE

Category	KPI	Target / Guard	Phase
Security	Audit findings (Critical/High)	0 open	P0–P1
Custody	Key rotation rehearsal	Completed; receipts	P1
Auction	Oversubscription ratio	≥ 1.2× (informational)	P2
Liquidity	Realized spread	Within policy bps	P2–P4
Liquidity	Depth ±1%	≥ floor	P2–P4
Treasury	Coverage-Months (CM)	≥ floor (e.g., 18m)	P2–P4
Supply	Burn yield (annualized)	Rising with adoption	P3–P4
Governance	Participation rate	≥ class quorum	P4
Identity	Liveness pass rate	≥ 98% (Open), ≥ 99% (KYC)	P1–P4
Operations	Incident count (S0/S1)	0 during P2–P3	P2–P3

10.4 D-30 → D+30 Timeline-Operational

Day	Activity	Primary Owner	Evidence
D-30	Final audit pass; lock parameter set v1	Security Lead	Auditor letters; Annex A hash
D-21	MPC ceremony; address registry freeze	Custody Lead	Ceremony + registry receipts
D-14	Auction dry-run; router parity test	Markets Lead	Dry-run receipts
D-7	Publish sale doc & addresses; start communications cadence	Comms Lead	Signed bulletin
D-1	Freeze window begins (code/params)	PMO	Freeze receipt

D0	Auction close; compute p*p^*p*; settle; seed/lock LP; mirrors live	Markets Lead	Settlement + lock receipts
D+1	KPI pack #1; venue health checks	Data Lead	D+1 report
D+7	Band density adjust (if hygiene pass); incident drill	Markets + Security	Rebalance + drill receipts
D+14	Governance cycle #1 opens	Governance Lead	Vote receipts
D+21	KPI pack #2; unlock calendar rehearsal	PMO + Vesting	Manifests
D+30	ORR checkpoint; parameter diffs (if any)	PMO	ORR receipt

10.5 RACI — Launch-Critical Workstreams

Workstream	**Responsible (R)**	**Accountable (A)**	**Consulted (C)**	**Informed (I)**
Contracts & Audits	Eng	Security	Auditors	PMO
Custody & MPC	Custody	Security	Eng, Legal	PMO
Auction Ops	Markets	PMO	Legal, Identity	Comms
Liquidity/Bands	Markets	PMO	Data	Comms
Identity Gateway	Identity	PMO	Legal	Security
Comms & Listings	Comms	PMO	Markets, Legal	All
Observability & IR	SRE	Security	Markets, Identity	PMO

10.6 External Validation and Listings

- **Exchanges/Aggregators:** Coordinate router parity and token metadata; no privileged routing.
- **Third-Party Dashboards:** Submit contract addresses and parameter registry hash for public tracking.

- **Public Archives:** Publish audit reports, SBOMs, and build attestations via signed repository releases.

10.7 Launch Communications Plan

- **Artifacts:** Sale overview, addresses, auction mechanics, governance primer, risk disclosures.
- **Cadence:** D-7, D-3, D-1, D0, D+1, D+7, D+21; each bulletin signed and archived.
- **Channels:** Official site, signed on-chain message, community forums, and verified social accounts.
- **Crisis Comms:** Pre-approved templates for S1/S0 incidents; 1-hour initial notice SLA.

10.8 Risk Register-Launch-Specific

Risk	**Likelihood**	**Impact**	**Early-Warning Signal**	**Mitigation**
Auction under-participation	Low–Med	Medium	Low unique bidders	Extend window; outreach; cap tweaks within bounds
Depth shortfall	Medium	High	Spread/Depth KPI breach	Reinforce bands; pause emissions; improve mirrors
Oracle freshness breach	Low–Med	High	Elevated quote age	Fail-closed; switch sources; publish status
Custody quorum outage	Low	High	Signer unavailability	Alternate quorum; emergency ceremony
Compliance reclassification	Low–Med	High	Jurisdiction bulletin	Mode switch to KYC; geo-gating; governance notice
Communications spoofing	Medium	Medium	Phishing reports	Single source of truth; signed bulletins; takedowns

Residual risk tracked in a color-coded matrix (likelihood × impact) with owners and review cadence.

10.9 Success Criteria and Post-Launch Reviews

- **Market Quality:** Depth ±1% ≥ floor; realized spread within policy for 14 consecutive days.
- **Solvency:** CM ≥ floor with no discretionary buybacks required.
- **Governance:** At least one completed C1/C2 cycle without execution errors.
- **Security:** Zero S0/S1 incidents; bug bounty triage SLA met.
- **Adoption:** UPI(30) trending upward; merchant retention ≥ target percentile.

Reviews

- **D+7 and D+21 Reports:** KPI dashboards, incident log (if any), parameter diffs.
- **D+90 ORR:** Formal sign-off to exit stabilization; backlog triage for V2 features.

10.10 Visual Index (Will be injected into the final report)

- **Fig. 10-A:** Phase roadmap (P0→P4) swimlane.
- **Fig. 10-B:** Go/No-Go gate checklist with artifact hashes.
- **Fig. 10-C:** D-30→D+30 timeline with freeze window.
- **Fig. 10-D:** RACI matrix heatmap.
- **Fig. 10-E:** Launch risk matrix and owners.
- **Fig. 10-F:** KPI scorecard template (pre/post TGE).

End Matter

Consolidated Actionable Recommendations

Policy & Monetary

- Finalize parameter registry v1.4 with hash commit; freeze TWAP 7,200s, freshness ≤120s, deviation fence 0.75%.
- Publish buyback controller receipts and hygiene dashboard; set alerting for fail-closed events.

Distribution & Vesting

- Publish unlock calendar with per-bucket addresses; enable ADV ($\kappa=0.05$) and depth ($\rho=0.20$) caps at the vesting gateway.
- Enable milestone verification playbooks for partner tranches; add clawback/expiry clauses to all open MoUs.

TGE & Liquidity

- Run auction dry-run; confirm ≥1.2× oversubscription in test; verify uniform-price settlement.
- Deploy primary LP band map around p*p^*p*; lock LP ≥12 months; seed mirrors and validate router parity.

Governance & Controls

- Activate C1/C2/C3 classes; enforce identity cap (0.1%); publish delegation expiry policy (90 days).
- Register emergency-pause TTL (24h) with receipts and bounded triggers.

HII & Missions

- Ship v2 template specs with weight vectors, SminS_{\min}Smin, and AFR targets; enable 3% baseline audit sampling with drift escalations.

- Roll out device cooldown (7 days) and geo-plausibility checks across templates.

Peace Store

- Launch fee band at 1.5% with discount cap ≤30% and budget BdB_dBd per epoch; enable T+2 settlement with 2% reserve default.
- Publish refund/chargeback SLAs and automatic policy responses.

Identity & Compliance

- Issue Passport VC with 180-day TTL; integrate selective disclosure; configure Open/KYC modes per region.
- Enable Sybil gate at ≤0.35 risk score; ring-fence anomalous clusters.

Security & Audits

- Complete dual audits; remediate to zero critical/high; publish attested SBOMs and build hashes.
- Confirm MPC quorums (3/5 routine; 4/7 constitutional); execute rotation ceremony and record receipts.

Observability & Reporting

- Publish monthly JSON manifests (parameters, KPIs, receipts); enable KPI alerts (depth ±1%, CM floors, FFR).
- Stand up status page with signed bulletins (D-7 → D+30 cadence).

Peace Soldiers — Annexes (EM-A … EM-K)

Annex Pack v1.4

This annex pack compiles EM-A through EM-K with tables and diagram placeholders ready for design layout.

EM-A — Parameter Registry (Single Source of Truth)

Canonical record of bounded parameters, defaults, ranges, and governance classes. Any change requires governance receipts and appears in monthly manifests.

Surface	Parameter	Default	Range / Bound	Class	Notes
Market Hygiene	TWAP window (s)	7,200	3,600–14,400	C2	Controller anchor
Market Hygiene	Freshness cap (s)	120	10–180	C2	Max quote age
Market Hygiene	Deviation fence (%)	0.75	0.25–2.00	C2	Abort if breached
Distribution	ADV cap κ	0.05	0.02–0.10	C2	Release $\le \kappa \cdot \mathrm{ADV30}$
Distribution	Depth cap ρ	0.20	0.10–0.30	C2	Release $\le \rho \cdot D \pm 1\%$
Treasury	CM floor (months)	18	9–24	C2	Solvency first
Governance	Identity cap (% of S)	0.10	0.05–0.20	C3	Per identity
Liquidity	Core / Near band shares	60% / 30%	Mapped	C2	Band density targets
Store	Fee (take rate)	1.5%	0.8–3.0%	C2	Category-aware
Store	Discount cap	30%	0–30%	C2	Budget-bounded

Figure F-1. Monetary Loop

EM‑B — Formula Library (Canonical Equations)

Supply identity: $S_{t+1} = S_t + M_t - B^{claim}_t - B^{bb}_t$

Claim-time burn: $B^{claim}_t = 0.0075 \cdot R_t$

Buyback sizing (guarded): $B^{bb}_t = \min(Surplus_t, HygienePass \cdot L_t)$

Unlock capacity: $A_t \le \min(\kappa \cdot ADV_{30,t}, \rho \cdot D_{\pm 1\%,t}, U^{schedule}_t)$

Slippage target: $\sigma(x) \approx x / D_{\pm w}$

HII score: $S = (\sum_k w_k q_k) \cdot (\alpha_g g + \alpha_t t + \alpha_u u) - \varphi$

Rate-limit credits: $c_i(t+\Delta) = \min(C, c_i(t) + \lambda\Delta)$

KPI exemplars: Burn yield = $(B^{claim} + B^{bb}) / S$; Clean GMV = Gross − refunds − chargebacks − fraud write‑offs.

Figure F‑2. Unlock Gantt with Caps and Gates

EM-C — KPI Dictionary (Definitions, Sources, Alerts)

KPI	Definition	Source	Frequency	Guard / Alert
Burn Yield	(B^{claim}+B^{bb})/S	Manifests, on-chain	Monthly	Rising with adoption
Depth ±1%	Aggregated depth at ±1%	Venue APIs	Daily	≥ floor; reinforce bands on breach
FFR	Controller freshness-fail rate	Controller logs	Daily	Near-zero; pause on breach
CM	Coverage-Months runway	Treasury ledger	Monthly	≥ floor
UPI(30)	Unique paying identities	Payments DB	Weekly	Upward trend
Top-10 Share	% held by top 10	Chain indexer	Weekly	≤ γ = 25%
Gini	Holdings inequality	Chain indexer	Monthly	< 0.60
Refund / Chargeback	Rates by category	Store ledger	Weekly	≤ policy thresholds

Figure F-11. KPI Scorecard Template

EM‑D — Receipt Schemas (Machine‑Verifiable)

Include policy_ref, timestamp, and signer identity. Hash over canonical field order.

buyback_execute: { action, epoch, twap_window_s, freshness_s, deviation_ok, notional, base_token, counter_token, tx_hash }

unlock_release: { action, epoch, bucket, schedule_id, calendar_due, released, caps{adv,depth,vol_gate}, carry_forward, tx_hash }

mission_claim_decision: { action, template_id, identity_hash, score_S, anomaly_A, decision, payout_pre_burn, payout_net }

order_settle: { action, order_id, buyer, merchant, notional, currency, discount, fee, reserve, payout, tx_hash }

passport_issue / passport_update: { action, did, vc_id/event, tier, ttl_days/risk_score, reason_code? }

gov_execute: { action, proposal_id, class, payload_hash, quorum, for_weight, against_weight, exec_block }

emergency_pause: { action, reason, ttl_h, signers[], tx_hash }

Figure F‑6. Monitoring Map — Signals and Alerts

EM-E — Governance Classes and Bounds

Class	Scope	Quorum	Timelock	Examples
C1 Routine	Non-privileged ops, reporting	10%	12h	Manifests, metadata rotations
C2 Parameterized	Bounded knobs (Annex A)	30%	48h	TWAP/freshness, band shares, κ/ρ
C3 Constitutional	Foundational changes	60%	96h	Identity cap, supply rules, mode flags

Proposals outside Annex A ranges are rejected at the gateway pre-vote. Delegations auto-expire after 90 days; per-identity cap 0.10% of supply.

Figure F-4. Governance Class Stack

EM-F — Address & Custody Registry

Post-audit publication; governance-gated modifications.

Role	Program	Address	Quorum	Rotation	Upgradeability
Treasury	Multisig/MPC	TBA	3/5	90d	N/A
Governance Gateway	Gov Proxy	TBA	4/7	90d	Guarded
Vesting	Escrow	TBA	3/5	90d	Guarded
Liquidity	LP Custody	TBA	3/5	90d	N/A

EM‑G — Auction Parameters (Discovery)

Parameter	Value	Notes
Mechanism	Uniform‑Price Auction	Commit‑reveal optional; sub‑window buckets
Window	24–72h	Time‑sliced buckets
Funding	SOL / USDC	Escrowed on commit
Per‑Identity Cap	Policy‑set	Gateway‑enforced
U_min	Published	Minimum unique participants
Settlement	Pro‑rata at p*	Surplus refunded
Compliance	Open/KYC by region	Mode flags constitutional

Figure F‑12. Auction Demand Curve — Clearing Price p*

EM-H — Consolidated Risk Matrix (Likelihood × Impact)

Category	Key Risk	Likelihood	Impact	Owner	Cadence	Primary Mitigation
Market & Liquidity	Depth shock at unlock	M	H	Markets	Weekly	ADV/Depth caps; deferrals
Monetary	Oracle/freshness failure	L–M	H	Treasury	Daily	Fail-closed; monitors
Governance	Capture/bribery	L–M	H	Governance	Monthly	Identity caps; quorums
Security	Key compromise	L	H	Security	Daily	MPC+HSM; rotation
HII	Fraud rings	M	H	Missions	Weekly	Graph analytics; Tier-2
Store	Chargeback surge	M	M–H	Commerce	Weekly	Reserves; SLA tightening
Compliance	Reclassification	L–M	H	Legal	Monthly	Mode switch; geo-gating

Figure F-10. Risk Matrix Heatmap — Placeholder

EM-I — Figure & Table Index (Production Mapping)

Figures

Tables

EM-J — Glossary & Acronyms

HII: Human-Impact Intelligence — evidence-based scoring of mission outcomes.

Clean GMV: Gross merchandise value net of refunds, chargebacks, and fraud write-offs.

CM: Coverage-Months — treasury runway metric.

UPI(30): Unique paying identities in trailing 30 days.

TWAP: Time-Weighted Average Price — market hygiene anchor.

FFR: Freshness-Fail Rate — share of epochs with stale quotes.

DLMM / CLMM: Programmable concentrated liquidity market makers.

C1/C2/C3: Governance classes — routine / parameterized / constitutional.

ADV30: 30-day average daily volume.

ρ / κ: Depth and volume caps for unlock control.

EM-K — Executive Synthesis

Design Objective: Align token value to verifiable human impact and real commerce through fixed supply, action-coupled burns, and surplus-driven buybacks under non-bypassable guards.

Market Path: Solana-native discovery via uniform-price auction; durable liquidity on programmable AMMs with LP locks ≥12 months; mirrors for routing parity.

Governance & Safety: Identity-bounded voting (0.1%), classed quorums/timelocks, bounded parameters, and fail-closed controllers. Audited contracts, MPC+HSM custody, reproducible builds, continuous monitoring.

Distribution: Community-major float (45%), runway-oriented treasury (25%), long-cliff team (15%), milestone-gated partners (10%), locked liquidity (5%). Unlocks paced by ADV/depth caps and volatility gates.

Economics: Monetary loop ties HII-verified claims to a 0.75% claim-time burn; surplus recycles to buybacks conditioned on hygiene tests. Target steady-state deflationary pressure ~2–5% annually in mature phases.

Transparency: Monthly signed JSON manifests publish parameters, KPIs, and receipts, enabling independent recomputation of supply, runway, and controller behavior.

Figure F-7. Launch Roadmap — P0→P4

Figure F-8. Store Checkout & Settlement Flow

Figure F-9. Identity Gateway Overview

Figure F-3. LP Band Map